



Autumn 2025

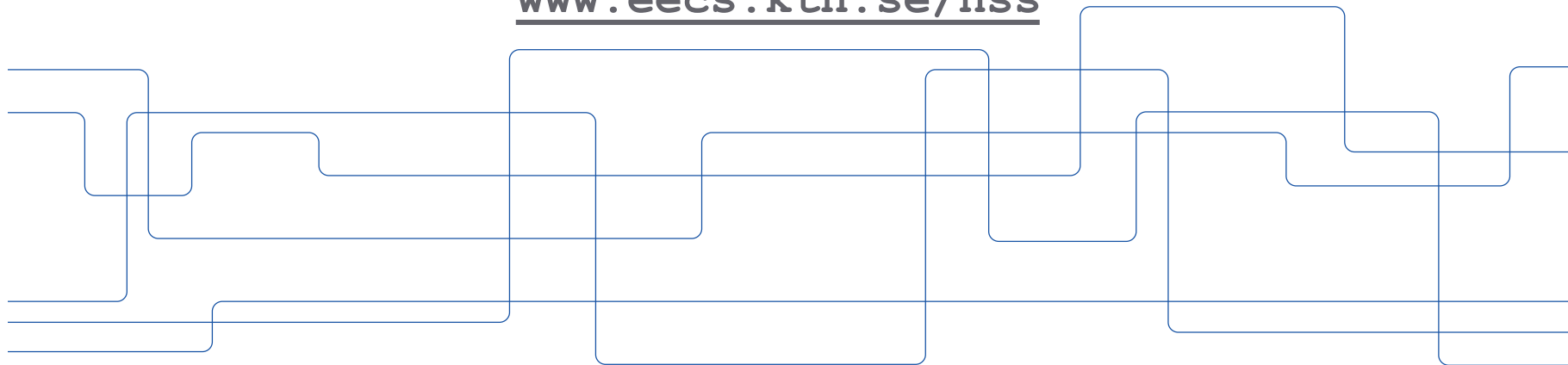
Secure Vehicular Communication Systems

Discussion leads

Kostas Kalogiannis
& Sheng Liu

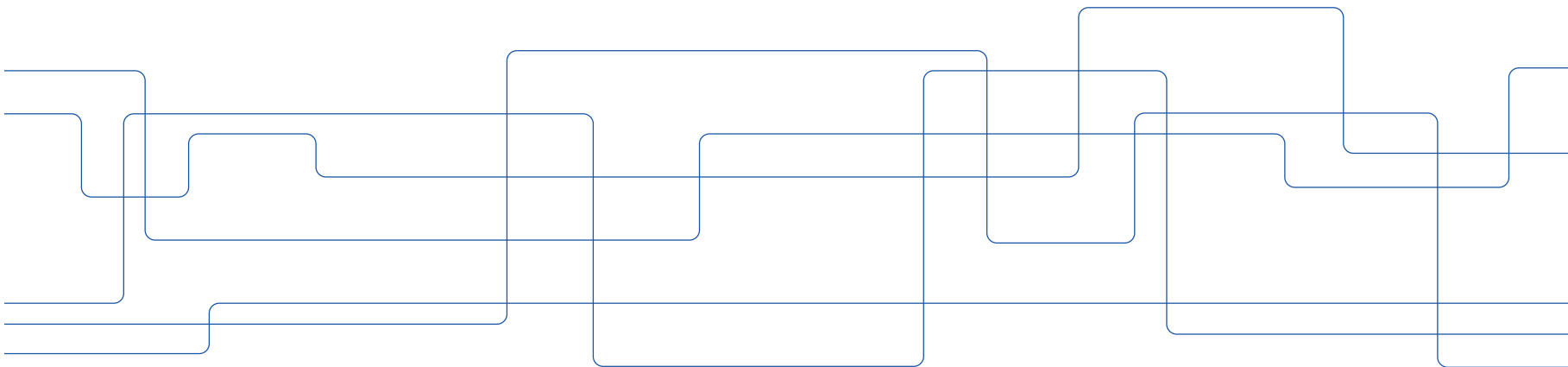
Panos Papadimitratos

www.eecs.kth.se/nss



Introduction

Background, challenge, requirements



Background

Vehicular Communication Systems (VCS)

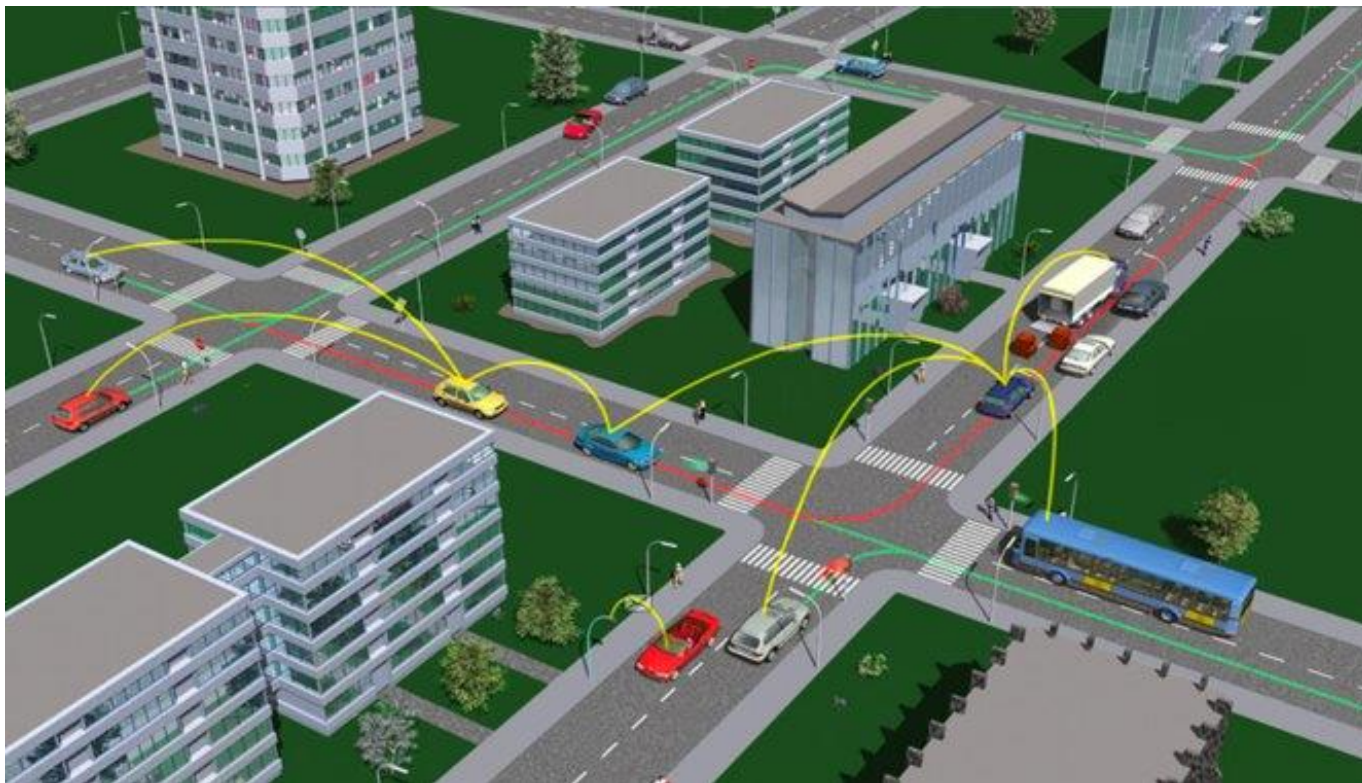
- Vehicles equipped with
 - Computers
 - Sensors
 - > Radar, lidar
 - > Cameras
 - > Positioning systems (GPS, Galileo)
 - Wireless Transceivers



Vehicle illustration courtesy of Daimler

Background (cont'd)

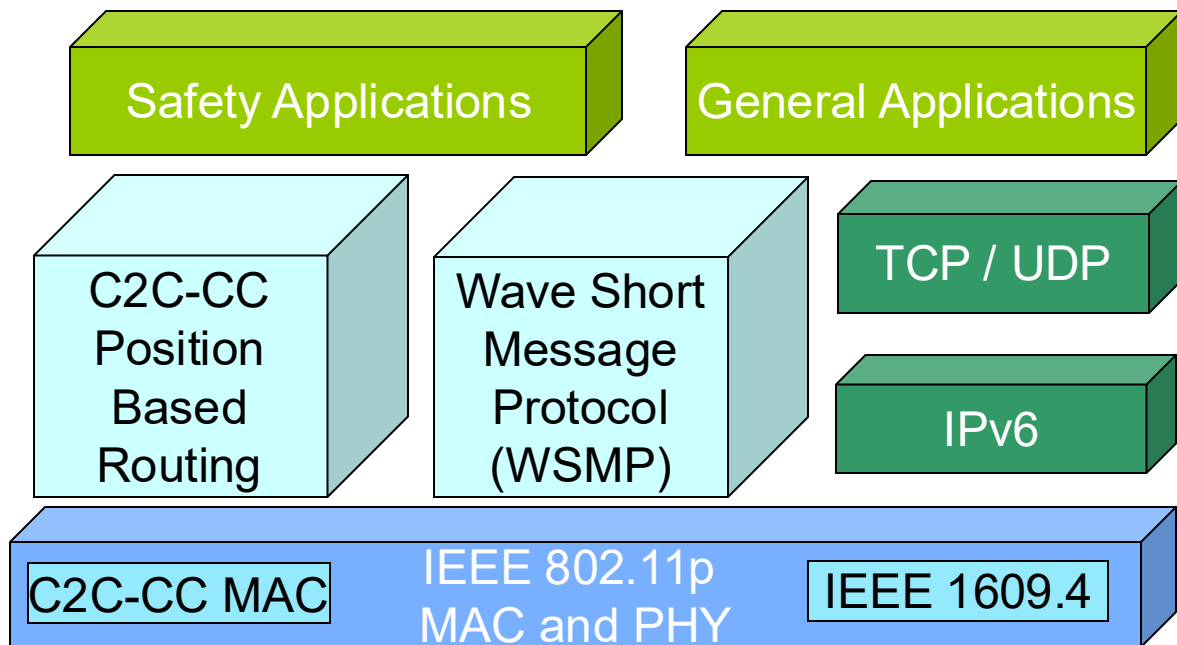
VCS



Background (cont'd)

VCS

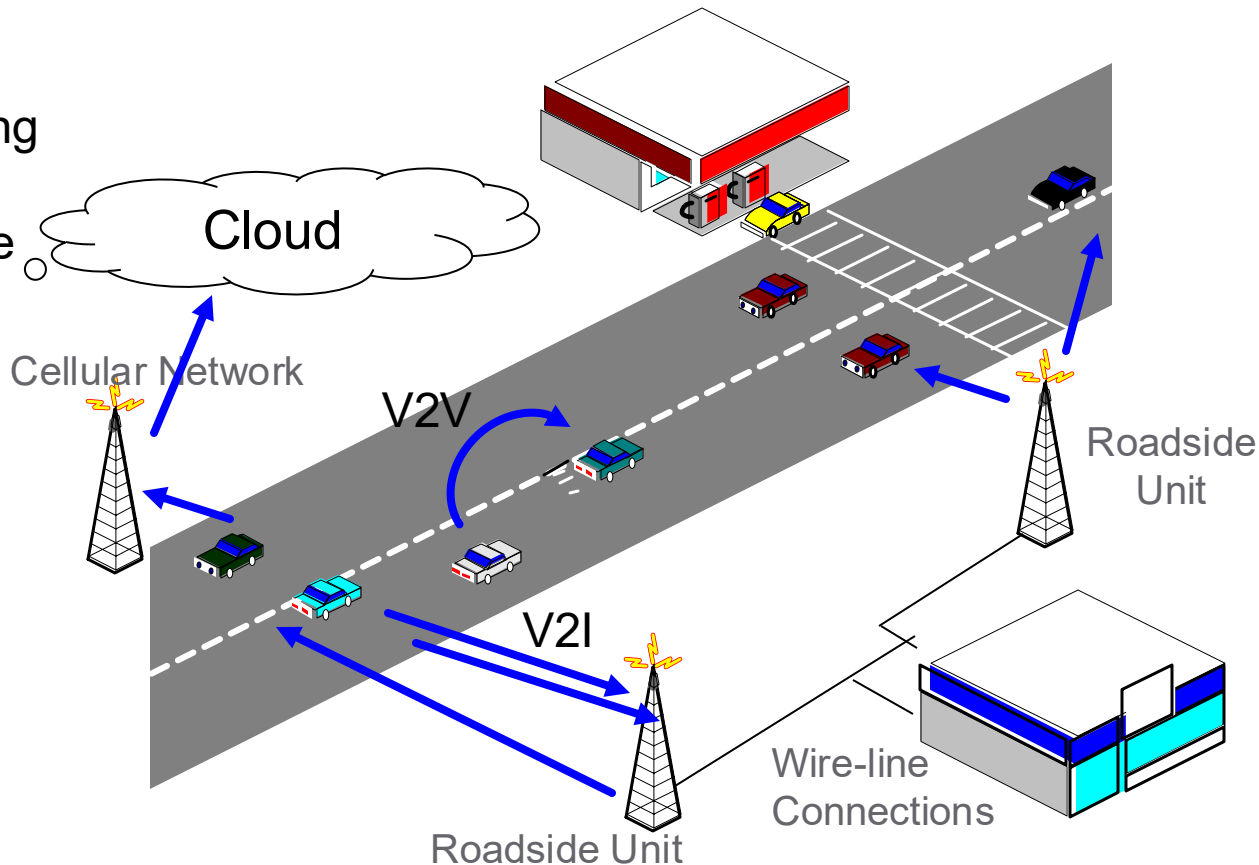
- VC-specific (e.g., safety) and TCP/IP communication



Background (cont'd)

V2V, V2I: V2X

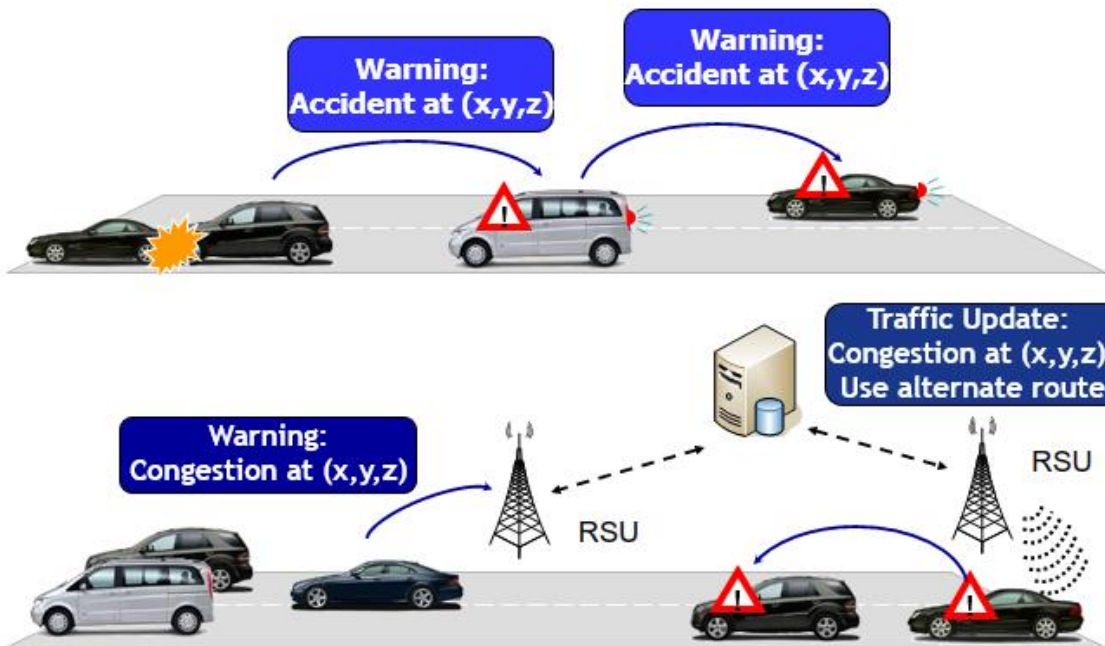
- V2X: Vehicle-to-everything
 - V2V: short range
 - V2I: short/long range
- Communication
 - ITS-G5 – 802.11
 - LTE-V2X – 4G/5G



Background (cont'd)

VCS Applications

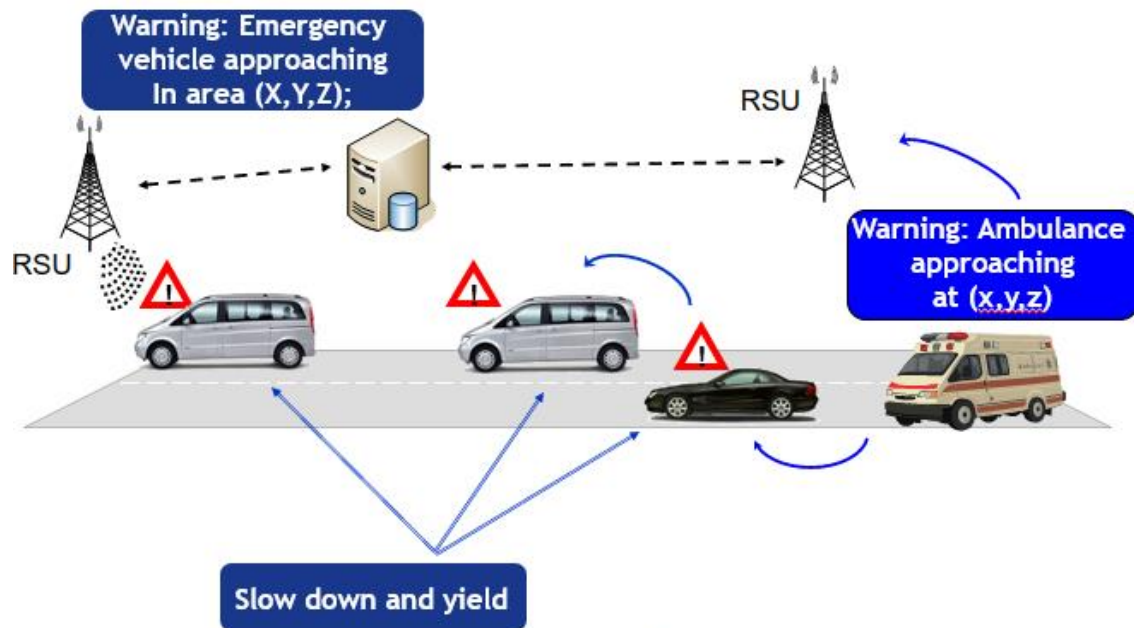
- Safety
 - Accidents Ahead
 - Unsafe road conditions
- Efficiency
 - Faster communication
 - Traffic updates



Background (cont'd)

VCS Applications

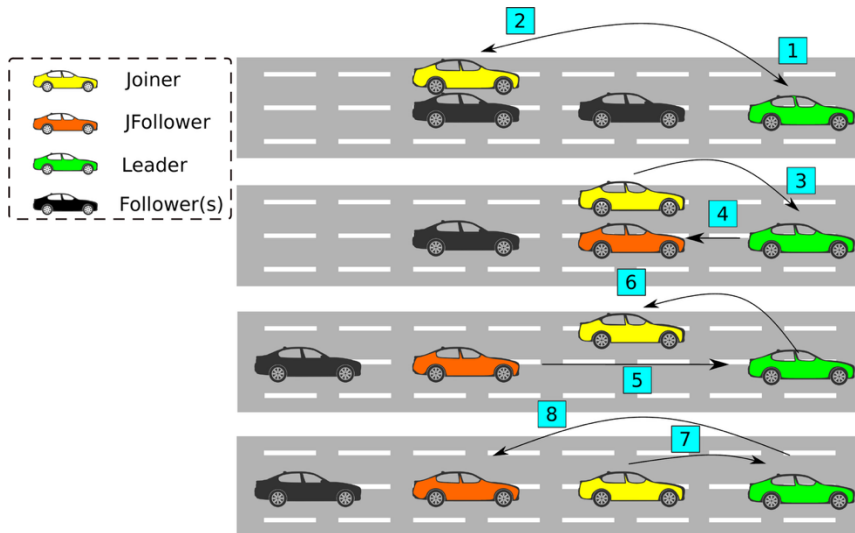
- Efficiency
 - Faster communication
 - Traffic updates
 - Emergency services



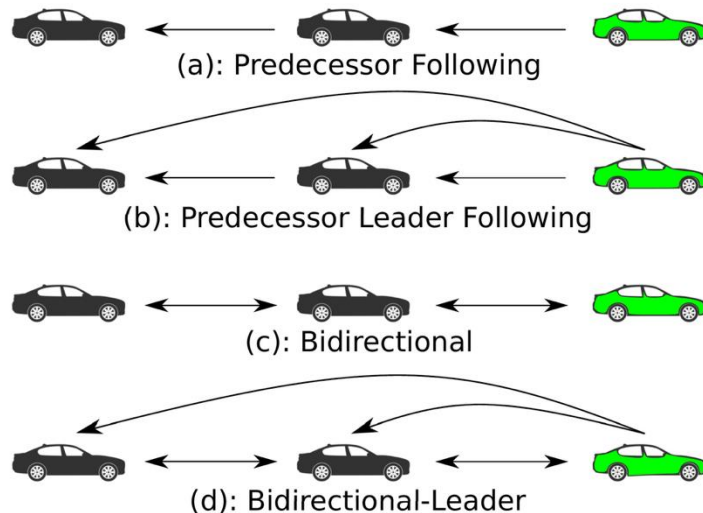
Background (cont'd)

VCS Applications

- Platooning
 - Autonomous Vehicles
 - Smaller gaps
 - Better fuel efficiency, road utilization



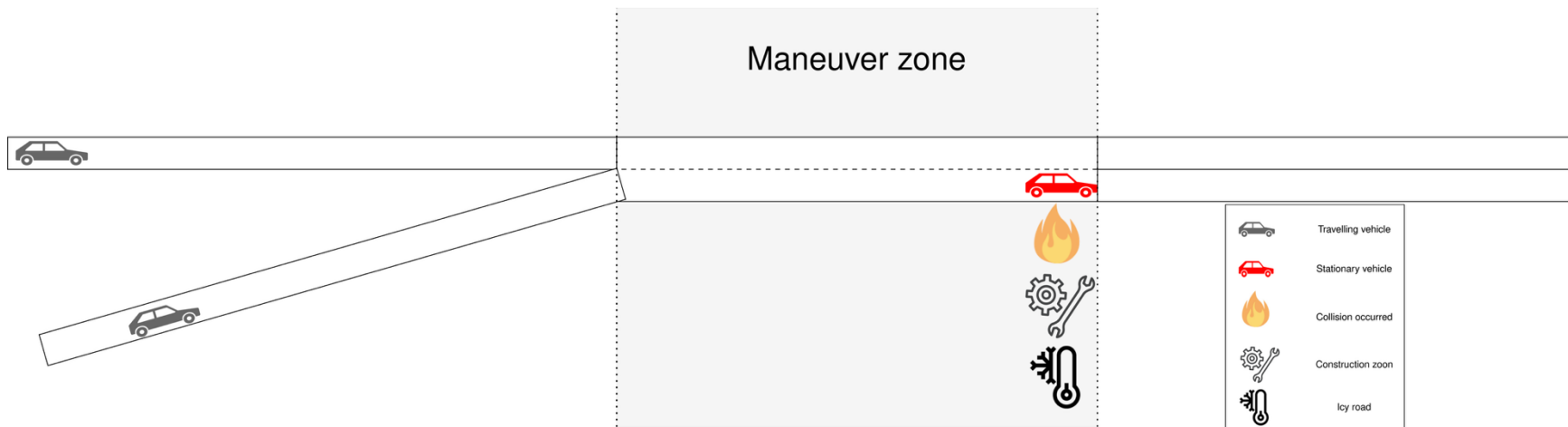
- Platooning network topologies
 - Leader position is significant
- Platoon maneuvers
 - Join/Exit



Background (cont'd)

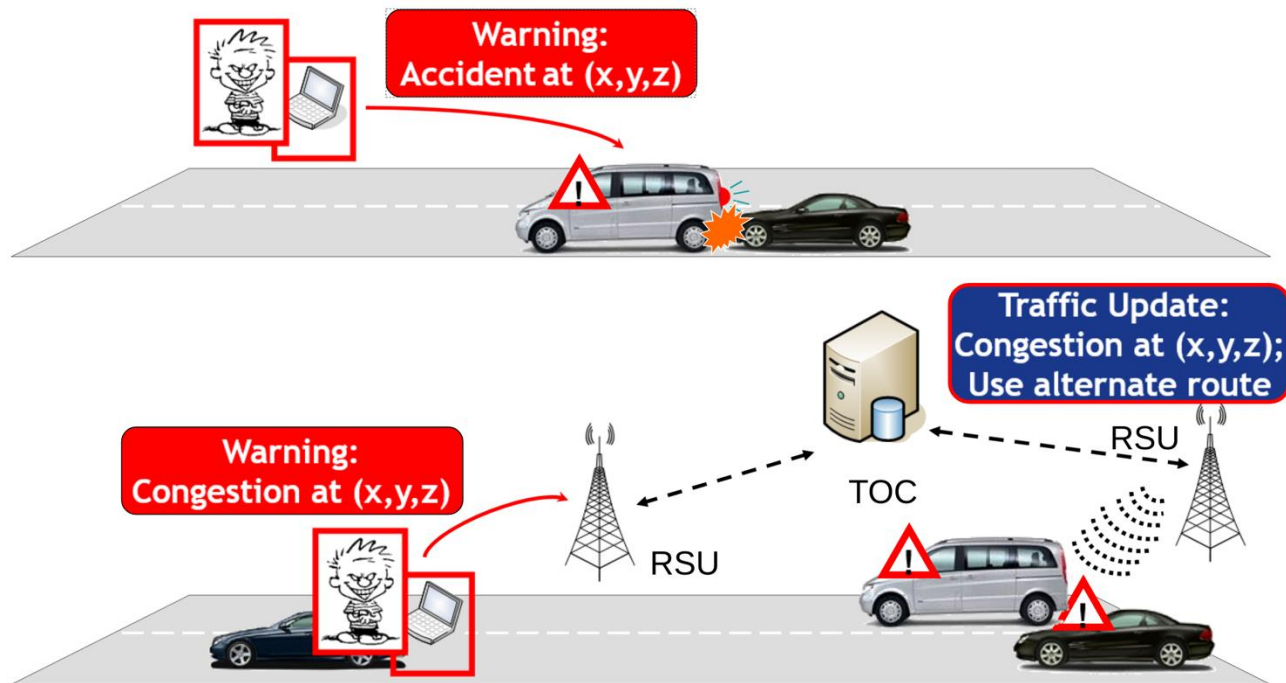
VCS Applications

- Coordinated Maneuvers
 - Help in transition areas (on/off ramps, intersections)
 - Mitigate slow human response



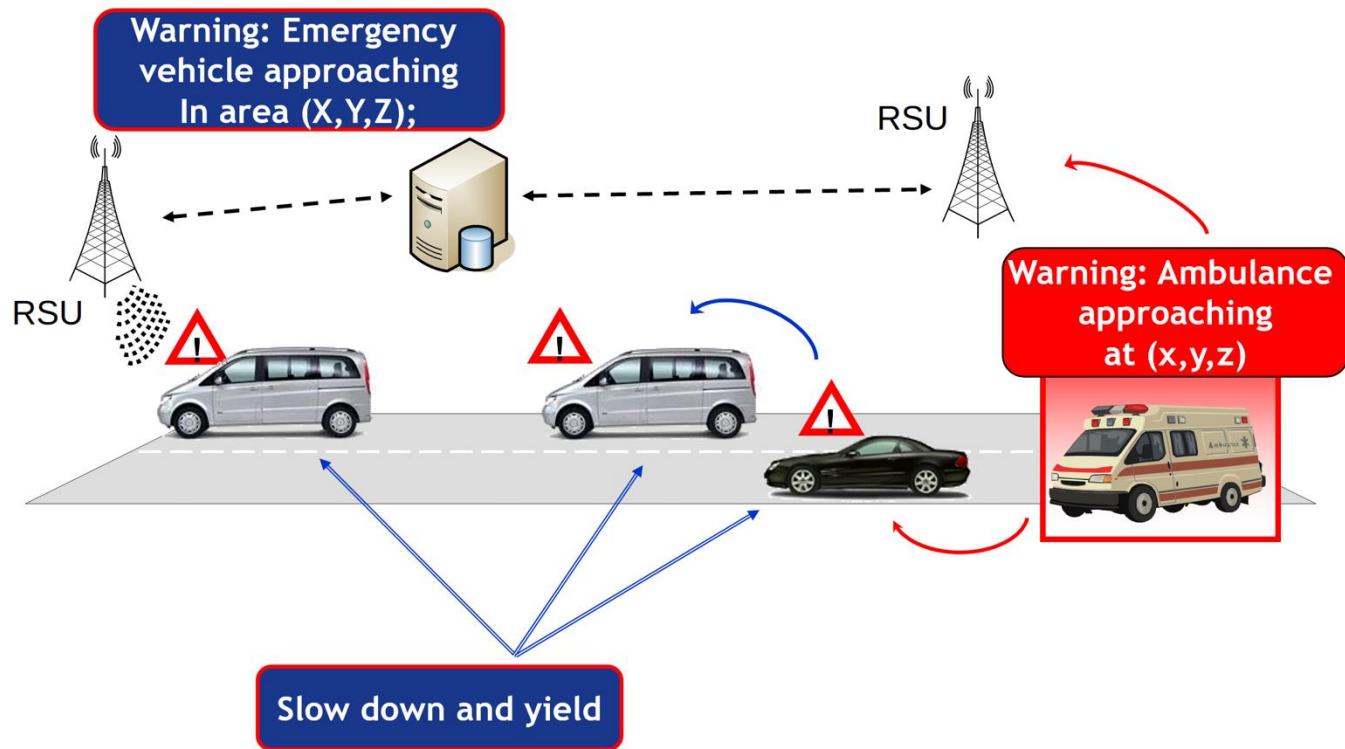
Attacking VCS

False data dissemination



Attacking VCS (cont'd)

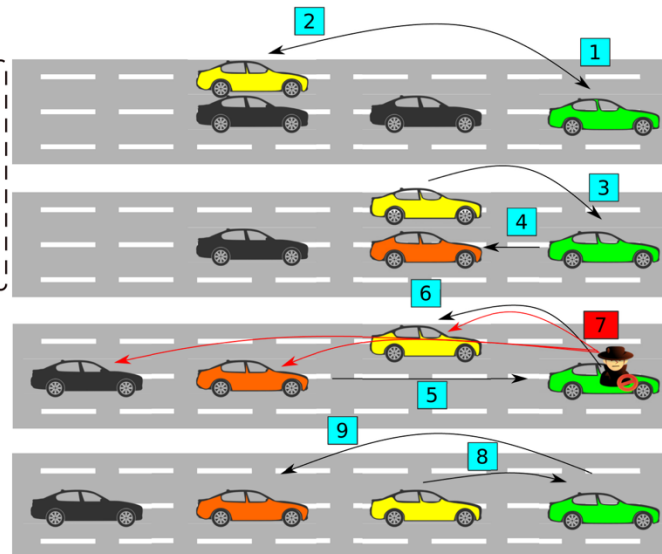
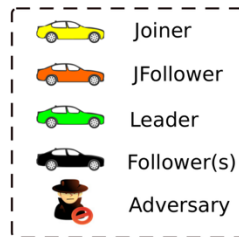
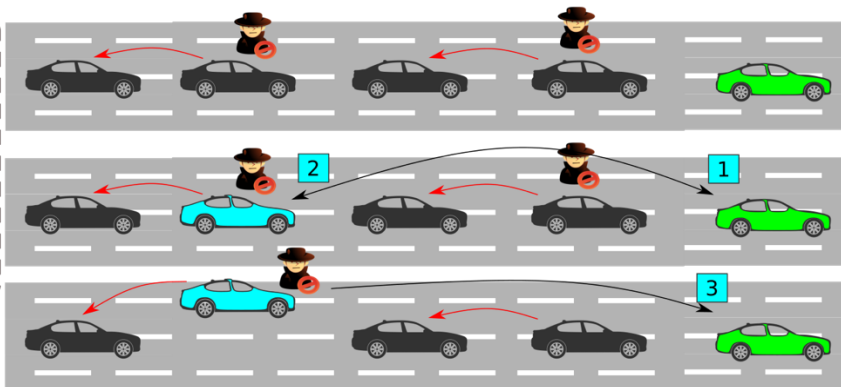
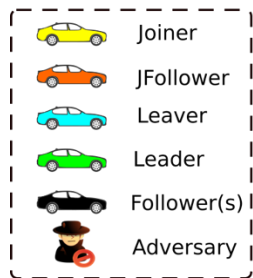
False data dissemination



Attacking VCS (cont'd)

False data dissemination

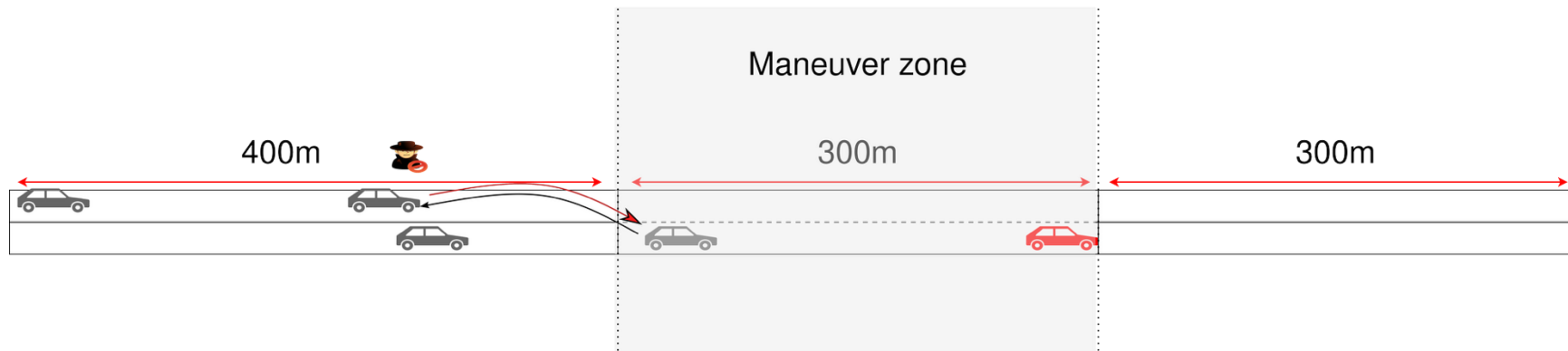
- Falsification during protocol execution
 - Complex protocols – multiple moving cars
 - Collusion!



Attacking VCS (cont'd)

False data dissemination

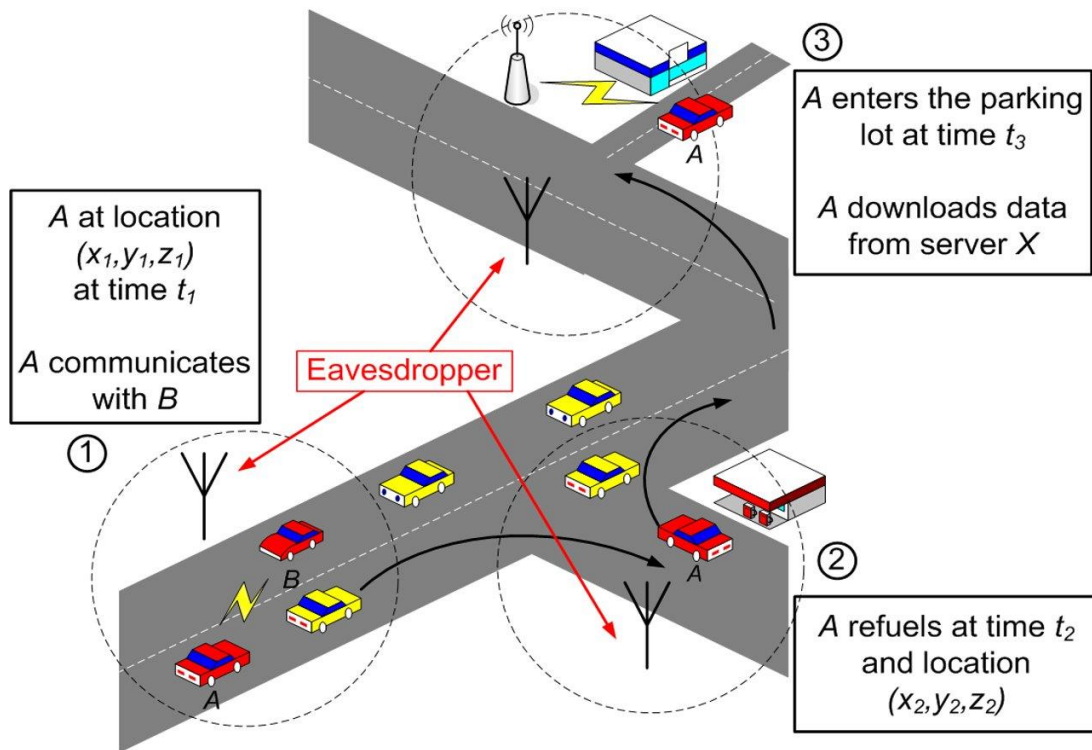
- Falsification during protocol execution
 - "I am next to you, wait!"
 - "I am further behind, make the maneuver!"



Attacking VCS (cont'd)

Privacy

- Vehicle authentication is required to ensure safety
- Vehicle actions are easily linked if a long-term identity of the vehicle is used (e.g., in a long-term certificate)
 - Linking of actions over a trip
 - Linking of trips over different days





Addressing the challenge

Security & Privacy (S&P) for VC – Why?

- Without robust designs, VC systems may facilitate antisocial behavior
- The deployment of vulnerable VC systems may cancel out their envisioned benefits
- Abused, poorly defended VC systems can cause damages and high cost
- Attackers and adversaries will always be present



Addressing the challenge (cont'd)

Why is VC S&P different?

- Complexity of the system
 - Hybrid (ad hoc, infrastructure) networking
 - Sensory inputs
- Tight coupling between users, applications, and network
- Pre-VC transportation systems and ‘legacy’ constraints and requirements
 - Liability identification
- Large scale and high mobility
- Strong privacy concerns

[PapadimitratosGH:C:2006]



Addressing the challenge (cont'd)

Objectives & Limitations

- Objectives
 - Secure communication
 - Anonymous and authentic communication
 - > *Baseline Privacy Enhancing Technology (PET)*
 - > *Messages generated by a given vehicle can be linked at most over a protocol-selectable period of time*
 - > *Shorter periods should lead to harder tracking*
 - Easy-to-implement
 - Low overhead
- Limitations
 - Communication cannot be regulated by the node/user
 - > Safety messaging will be essentially an 'always-on' application
 - Vehicle-originating wireless transmissions are easy to eavesdrop
 - > *Data link very similar to a widely adopted technology: IEEE 802.11*
 - > *Very large and increasing numbers of 802.11 access points already deployed*
 - > *Road-side infrastructure deployed for other services could be subverted into acting as an eavesdropper*

Addressing the challenge (cont'd)

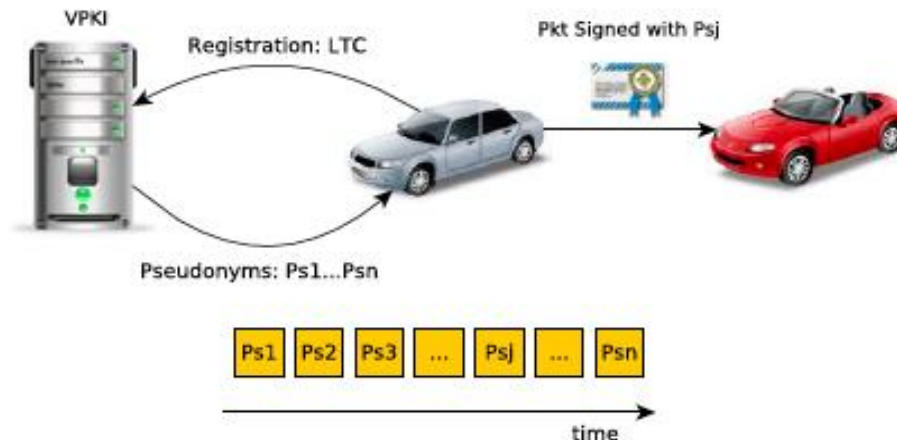
VCS S&P

- Basic requirements

- Authentication and integrity
- Non-repudiation
- Authorization and access control
- Anonymity (conditional)
- Unlikability (longer-term)
- Accountability

- Vehicular Public-Key Infrastructure (VPKI)

- Ephemeral pseudonymous credentials
- Long-term credentials





Addressing the challenge (cont'd)

ETSI Technical Specification 102 941

- ISO/IEC 15408-2 identifies 4 key attributes
 - Anonymity – insufficient alone as a solution
 - Pseudonymity - accountable use a resource or service without disclosing identities
 - Unlinkability - multiple uses of resources or services without being linked
 - Unobservability - use a resource or service without others, especially third parties, being able to observe
- ITS Privacy is provided in two dimensions
 - privacy of ITS registration and service authorization
 - privacy of communications between ITS-Ss

Source: https://www.etsi.org/deliver/etsi_ts/102900_102999/102941/01.04.01_60/ts_102941v010401p.pdf



Addressing the challenge (cont'd)

IEEE 1609.2™-2016

- Secure Protocol Data Unit (PDU) format: signed data, encrypted data
- Certificate format:
 - Certificates for signing application PDUs: pseudonymous (no identification of sender) and identified
 - Certificates for Certificate Authorities (CAs)
- Certificate revocation list (CRL) format
- Peer-to-peer certificate distribution to allow devices to “learn” unknown certificates

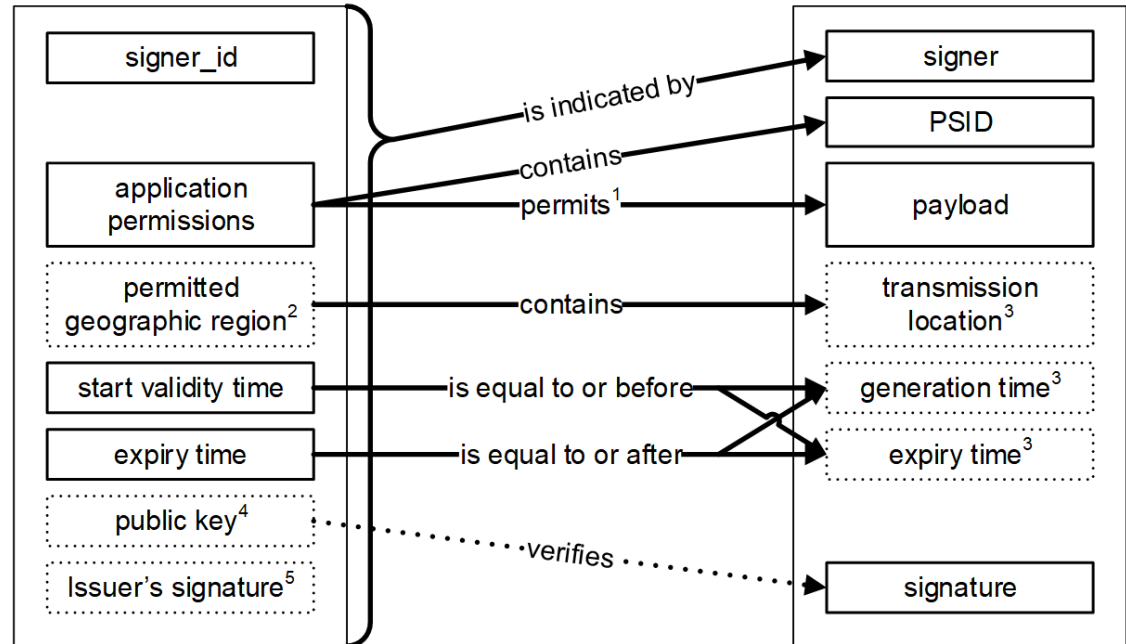
Addressing the challenge (cont'd)

IEEE 1609.2™-2016 (continue)

- Consistency between PDU and certificate

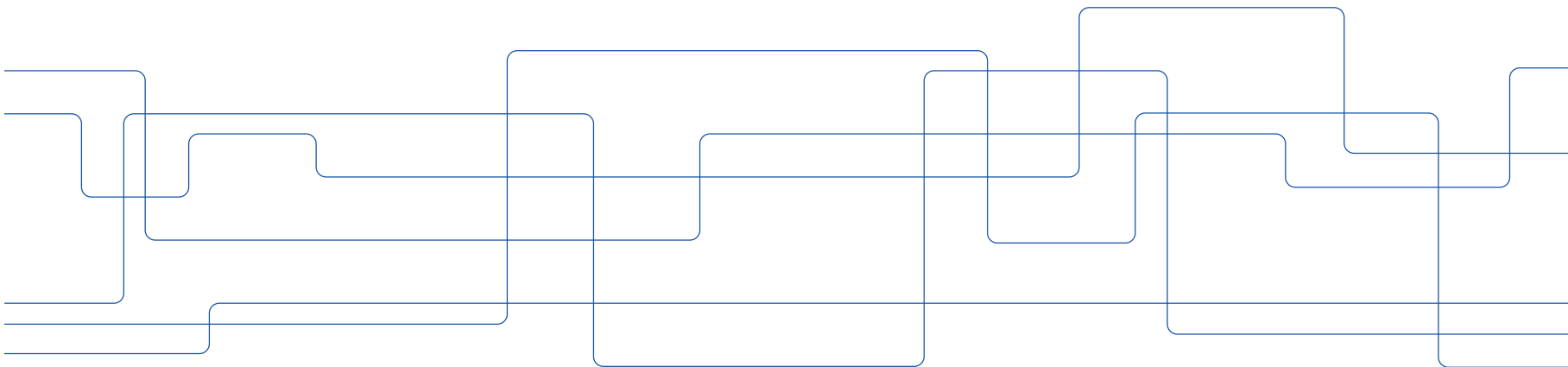
Signing Certificate

Signed PDU



Secure VCS

Basic components and performance





Secure VCS

Pseudonymous authentication

- Pseudonym
 - Remove all identifying information from certificate
- Pseudonym format

PNYM

PSNYM-Provider ID	PSNYM Lifetime
Public Key K_i	
PSNYM-Provider Signature	

Short-term Cert()

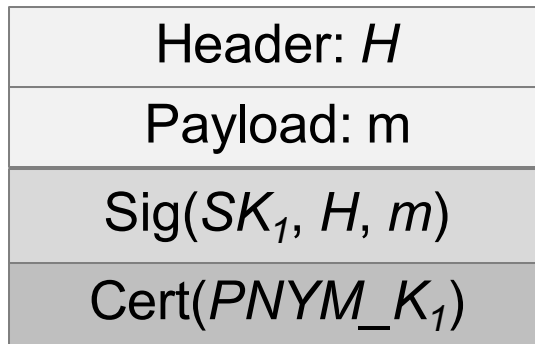
- Pseudonym provider: a trusted third party

Secure VCS (cont'd)

Pseudonymous authentication

CAM packet

1. Generate signature with SK_1
2. Append certificate
3. Send packet



1. Validate certificate (if not previously done so)
2. Validate signature
3. Validate geo-stamp in the header
4. Accept/Reject packet

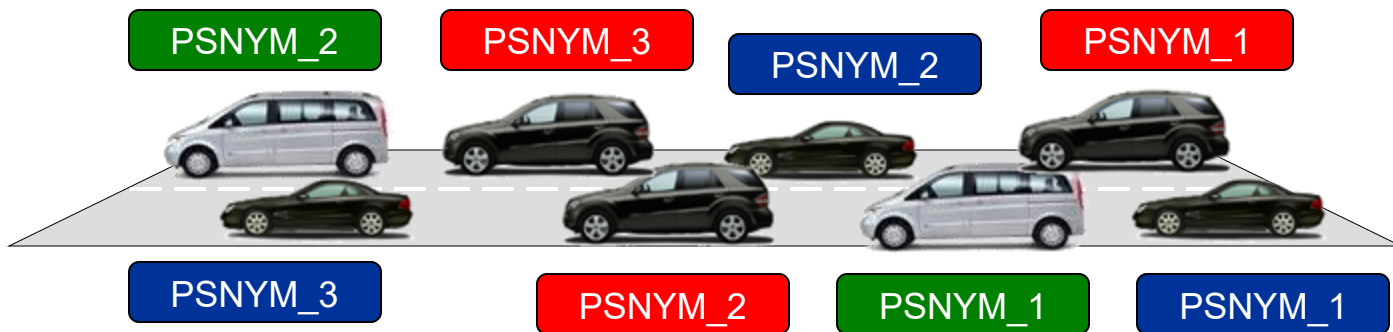


- Vehicle-to-Vehicle (V2V)/Vehicle-to-Infrastructure (V2I) (V2X) message communications are digitally signed
- Messages are signed with the private key corresponding to the currently valid pseudonym
- Cryptographic operations in a Hardware Security Module (HSM)

Secure VCS (cont'd)

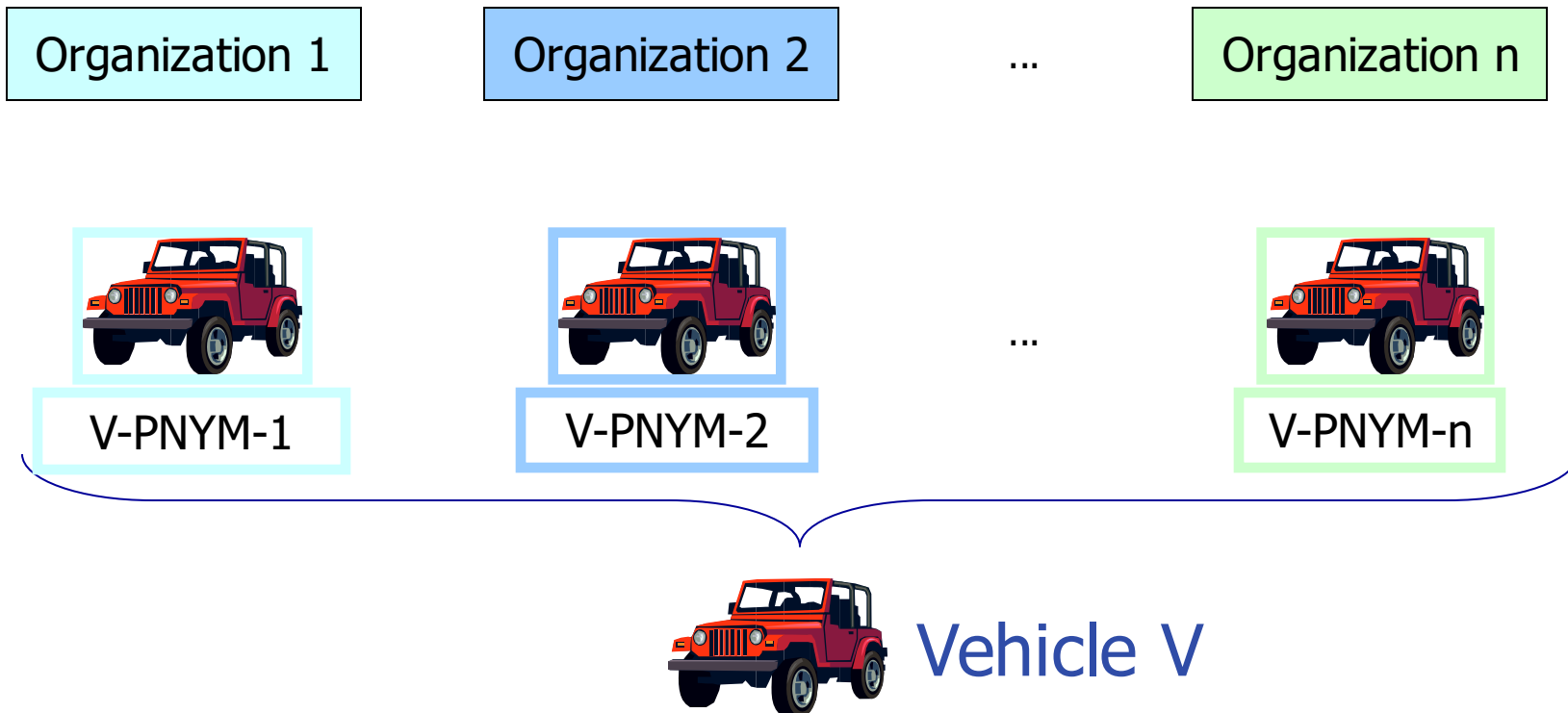
Pseudonymous authentication

- Equip vehicles with multiple pseudonyms
 - Alternate among pseudonyms over time (and space)
 - Sign message with the private key corresponding to pseudonym
 - Append current pseudonym to signed message



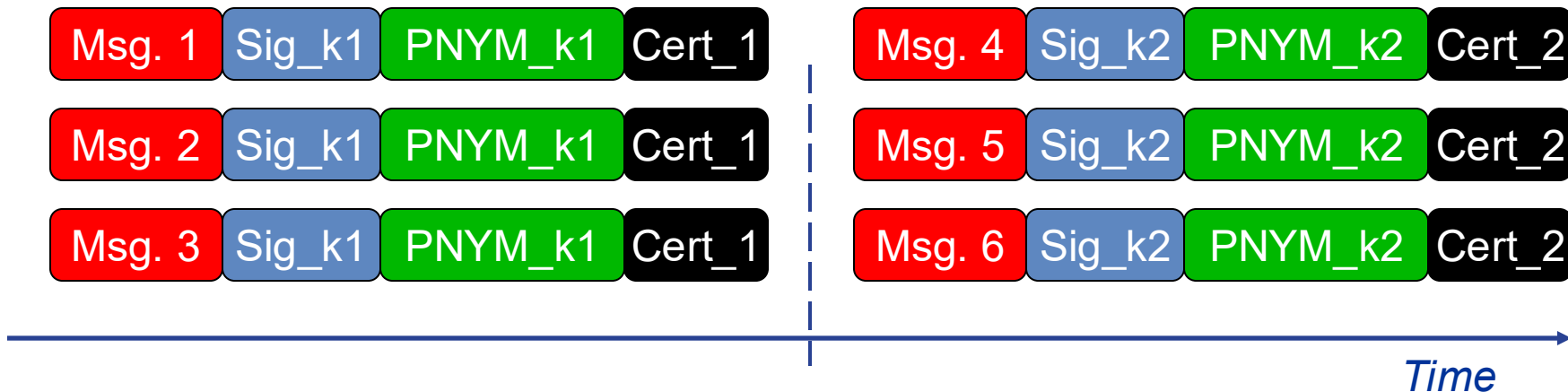
Secure VCS (cont'd)

Pseudonymous authentication



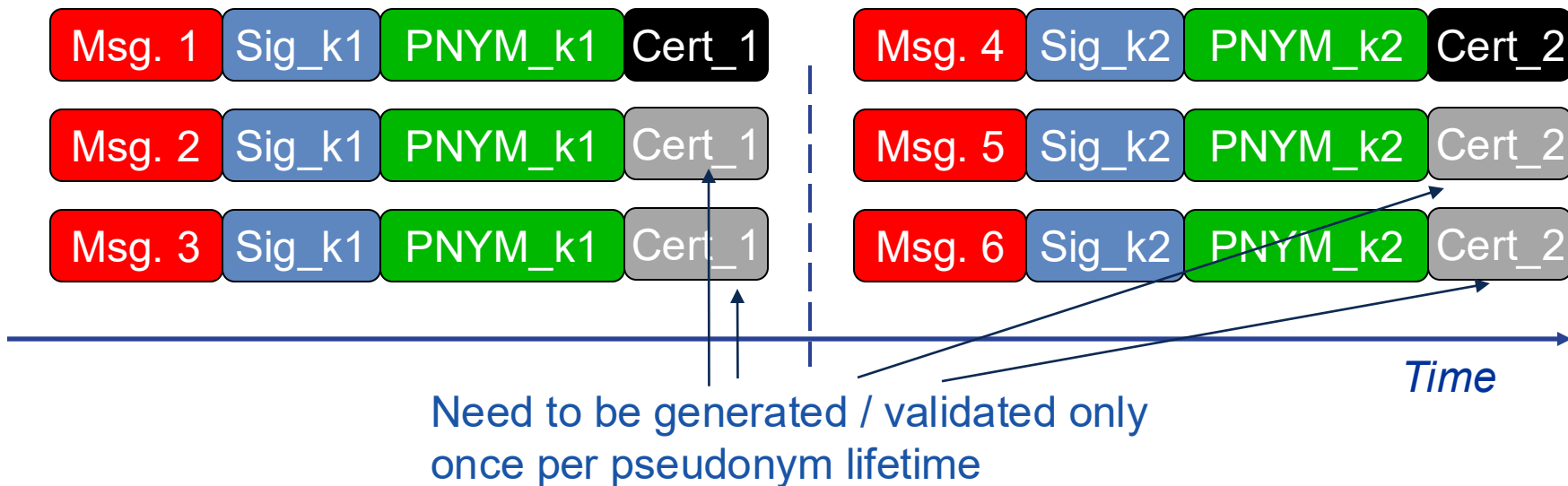
Secure VCS Cost

Optimization 1



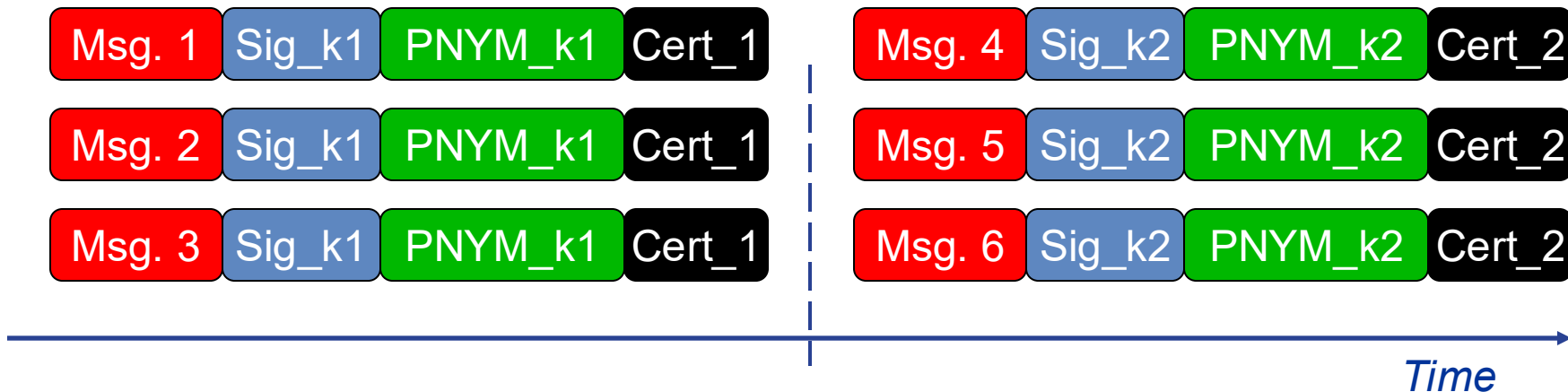
Secure VCS Cost (cont'd)

Optimization 1



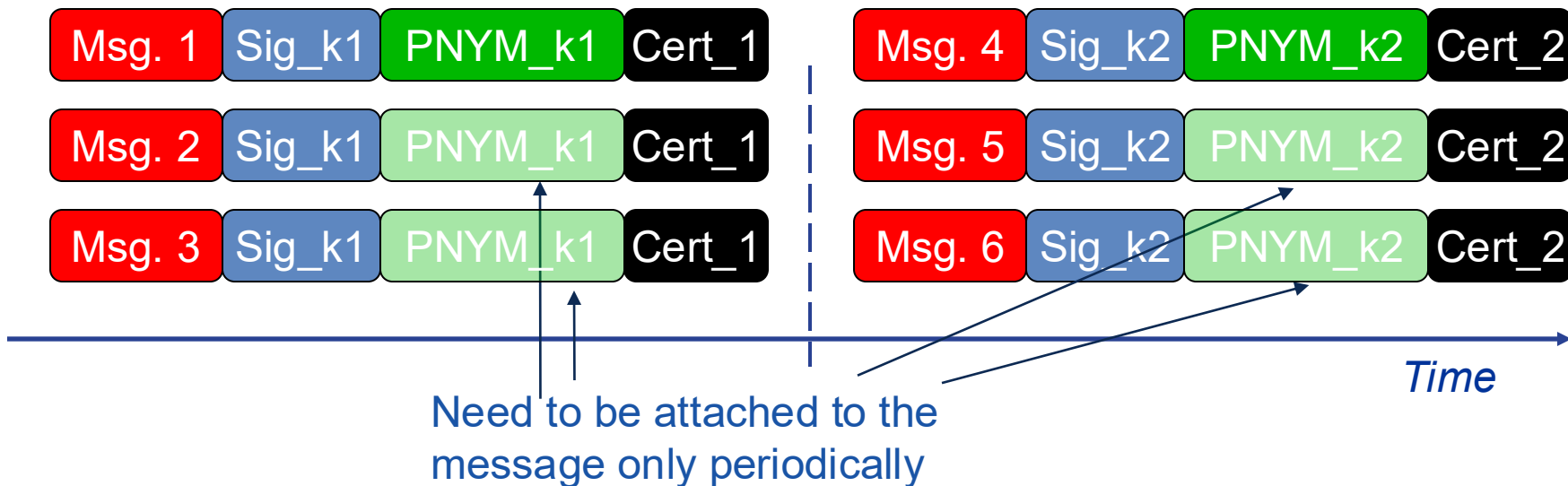
Secure VCS Cost (cont'd)

Optimization 2



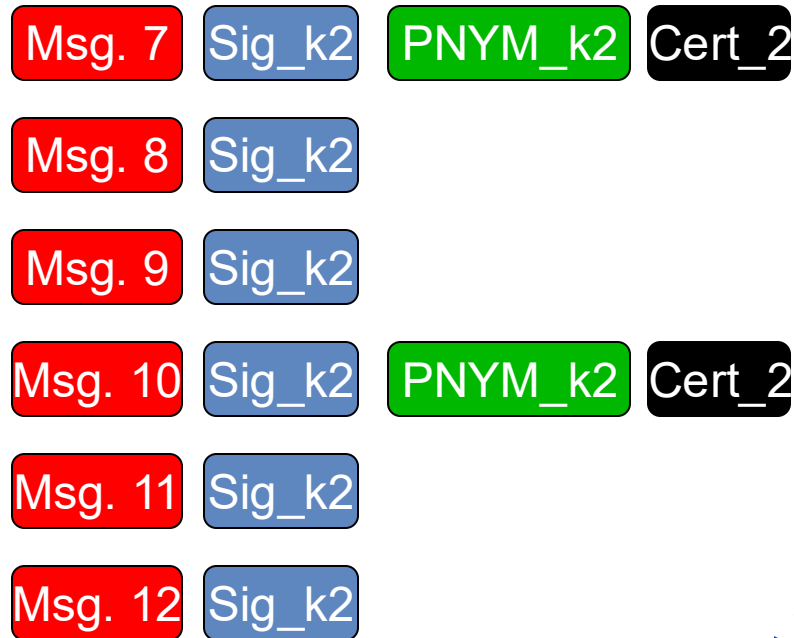
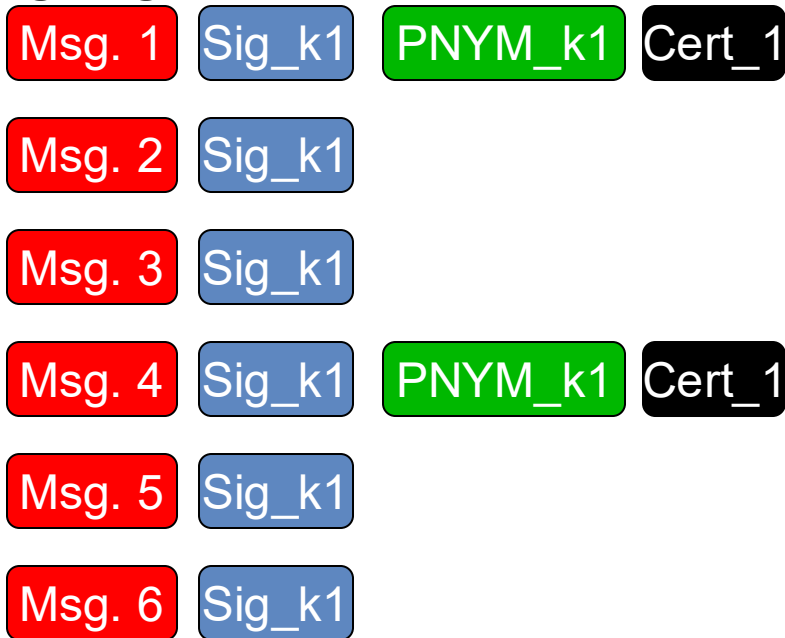
Secure VCS Cost (cont'd)

Optimization 2



Secure VCS Cost (cont'd)

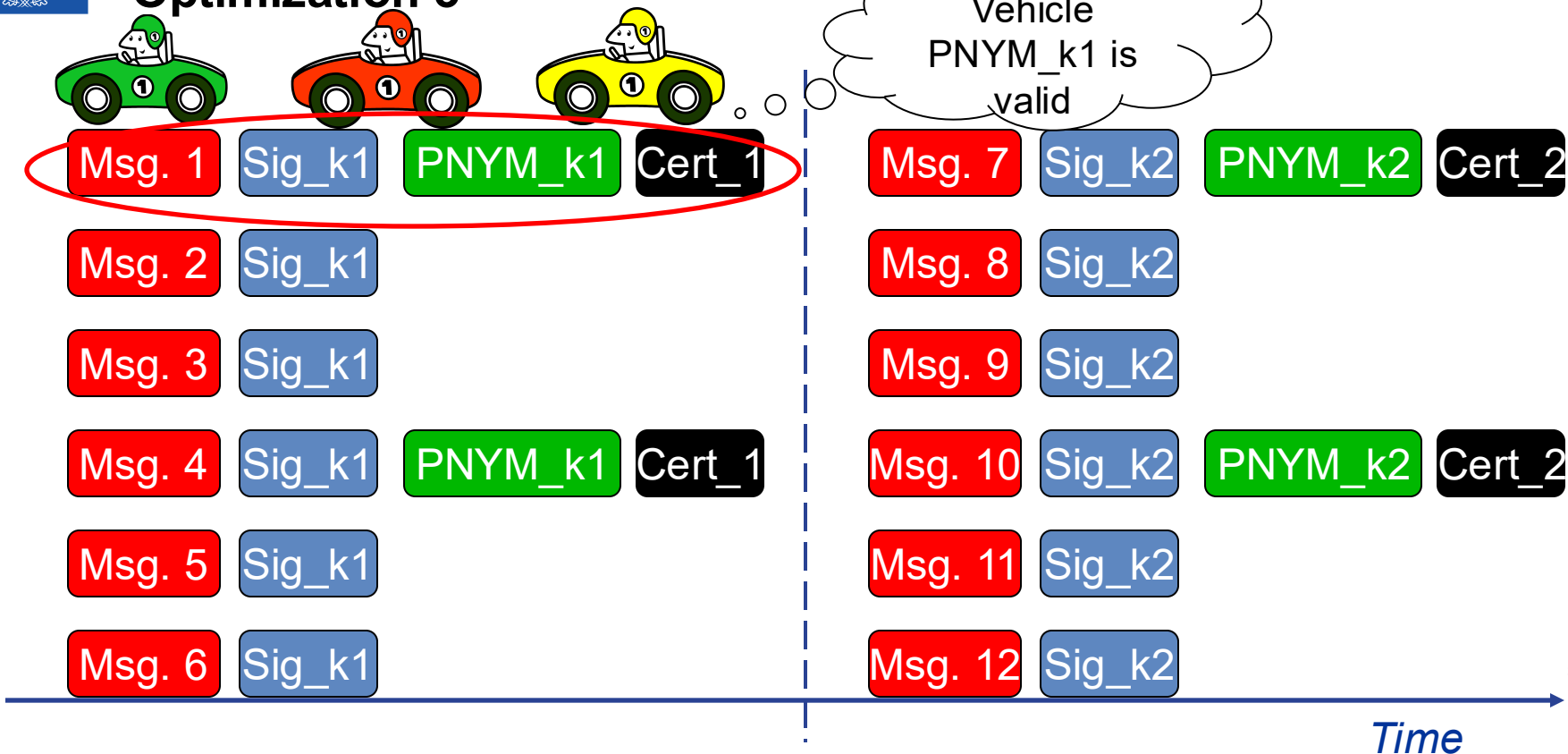
Optimization 2



Time

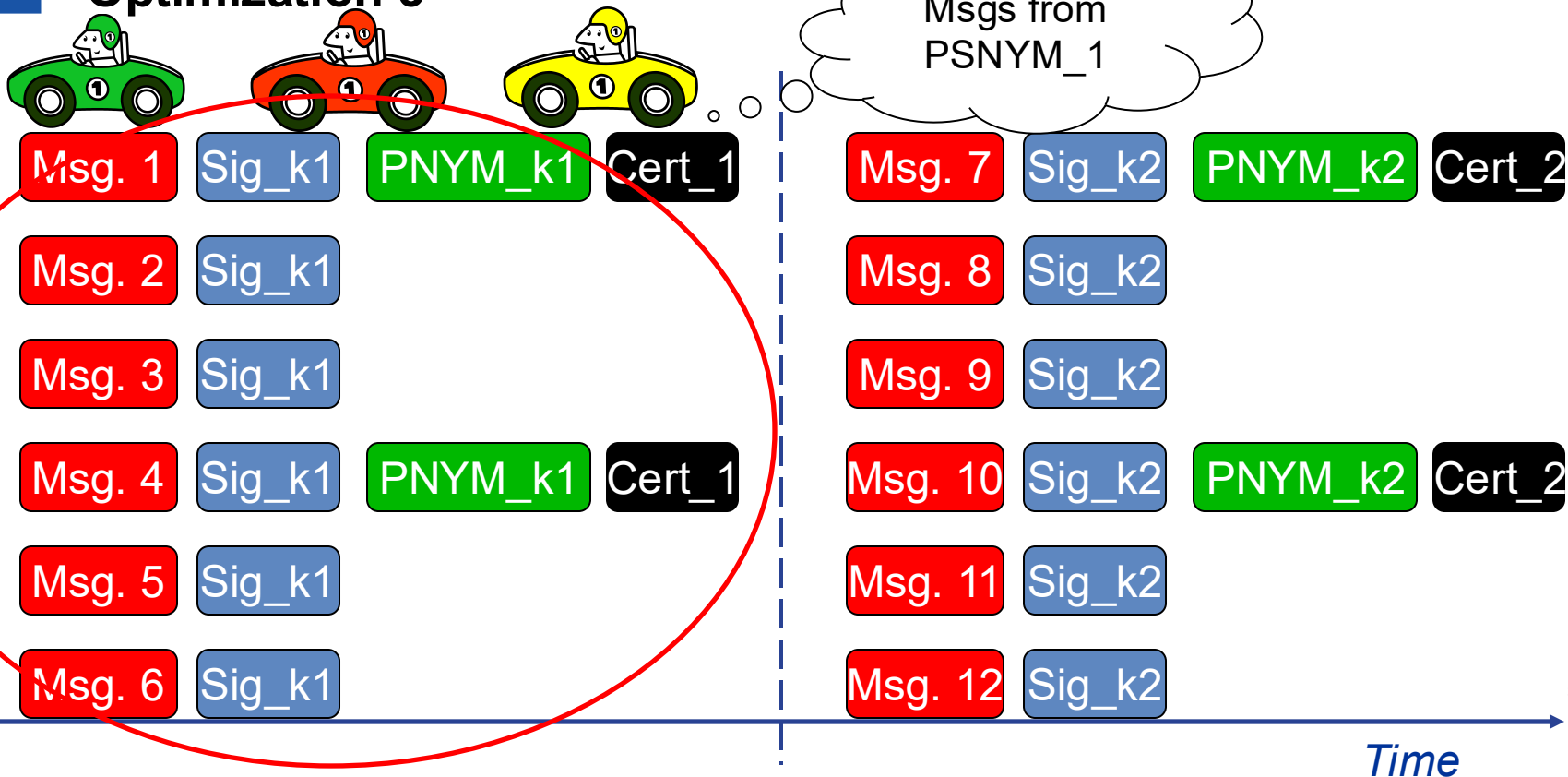
Secure VCS Cost (cont'd)

Optimization 3



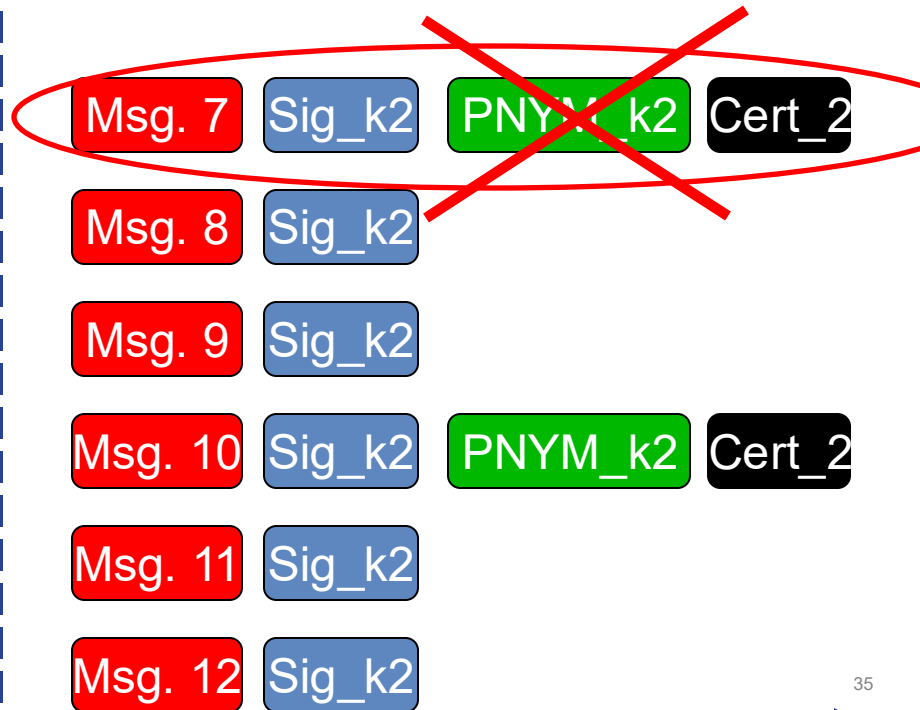
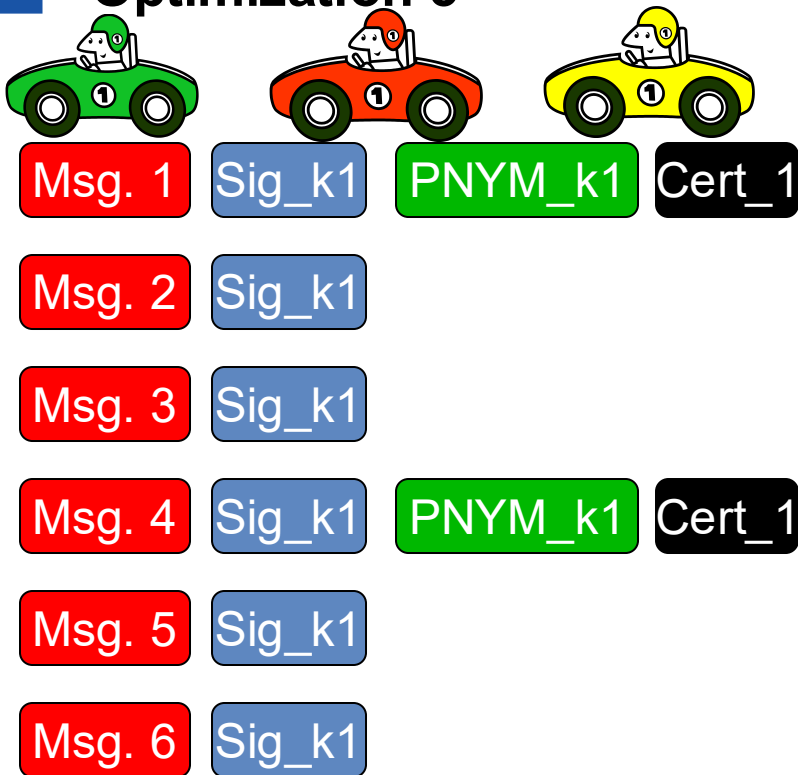
Secure VCS Cost (cont'd)

Optimization 3



Secure VCS Cost (cont'd)

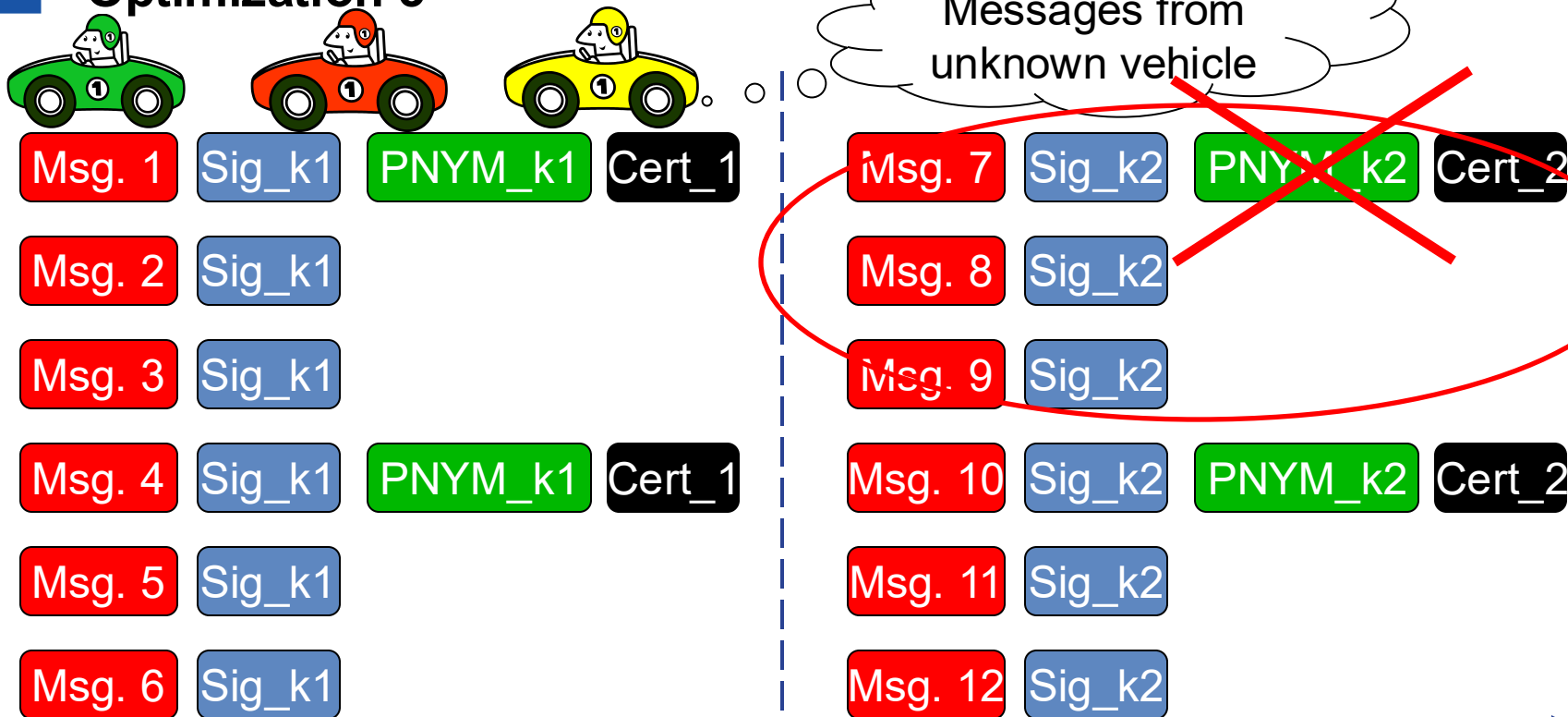
Optimization 3



Time

Secure VCS Cost (cont'd)

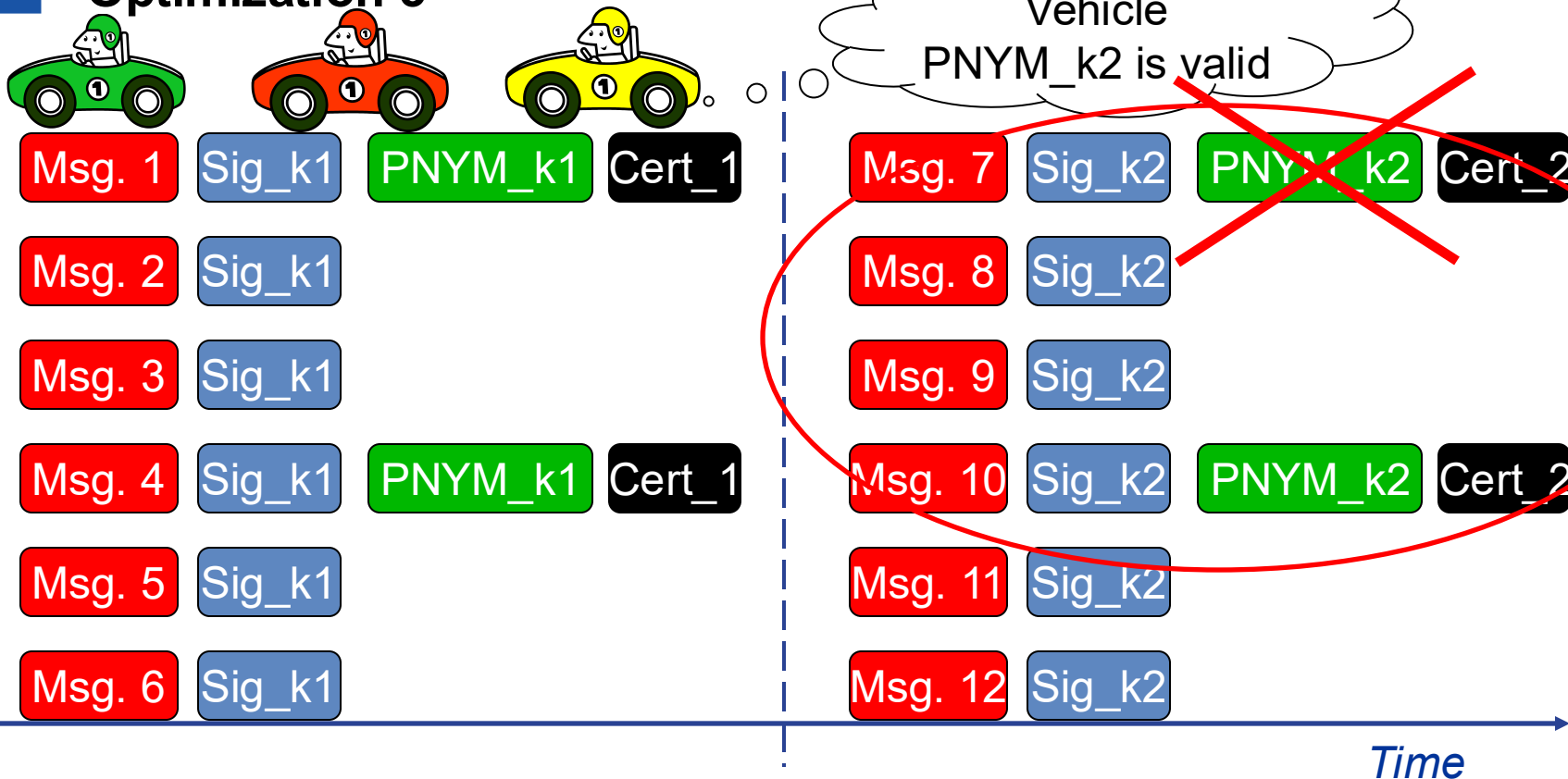
Optimization 3



Time

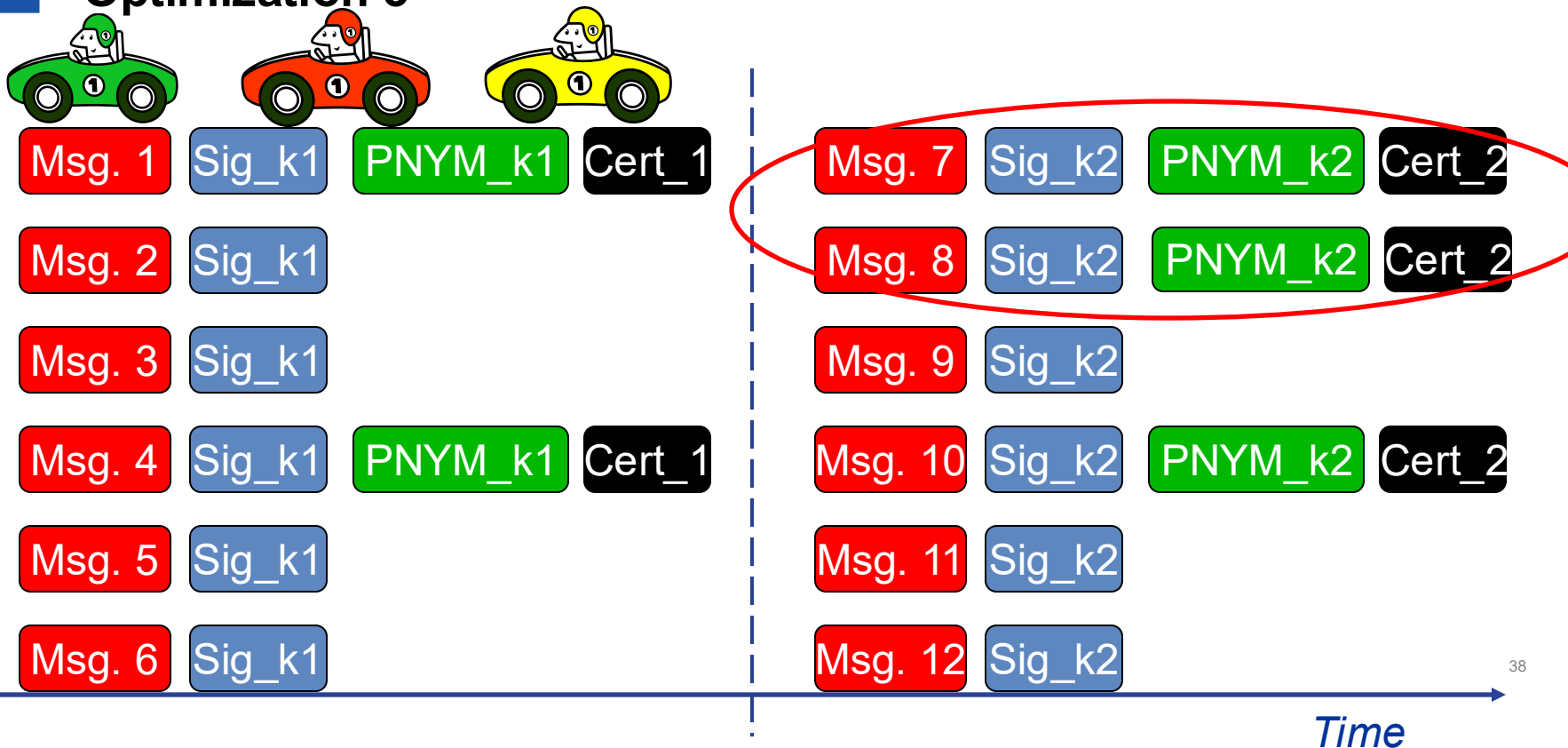
Secure VCS Cost (cont'd)

Optimization 3



Secure VCS Cost (cont'd)

Optimization 3



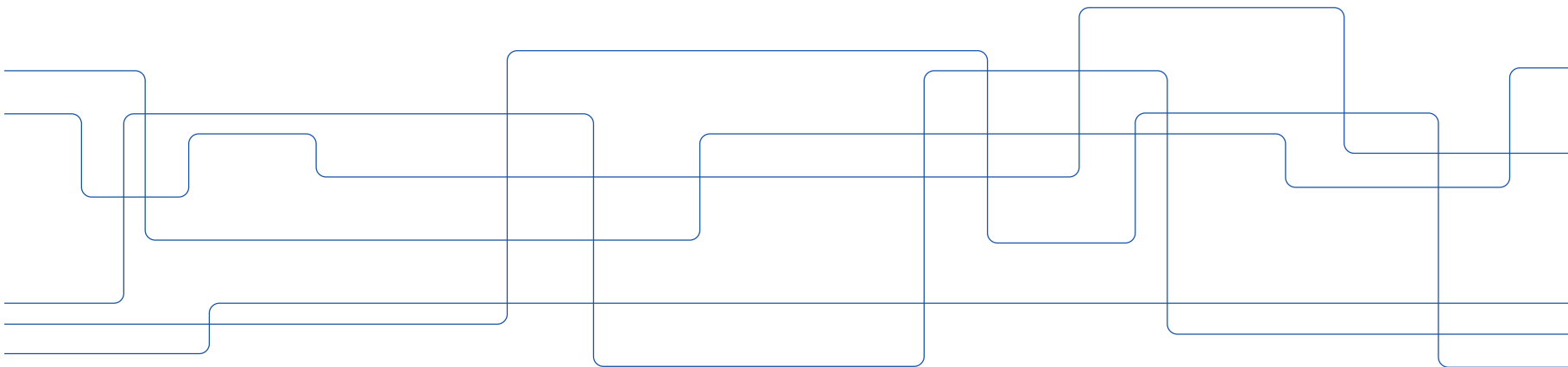


Secure VCS Cost (cont'd)

Recap

- Certify and validate a pseudonym once during its lifetime
- Append the certificate one every α messages
 - $\alpha = \text{certificate period}$
- When a new pseudonym is issued, transmit a certificate for p consecutive messages
 - $\beta = \text{push period}$

Thwarting attacks



Attack Detection/Mitigation

Thwarting clogging DoS against VCS

Limited OBU resources

- E.g., 4 msec/verification >> max. 250 signatures/sec

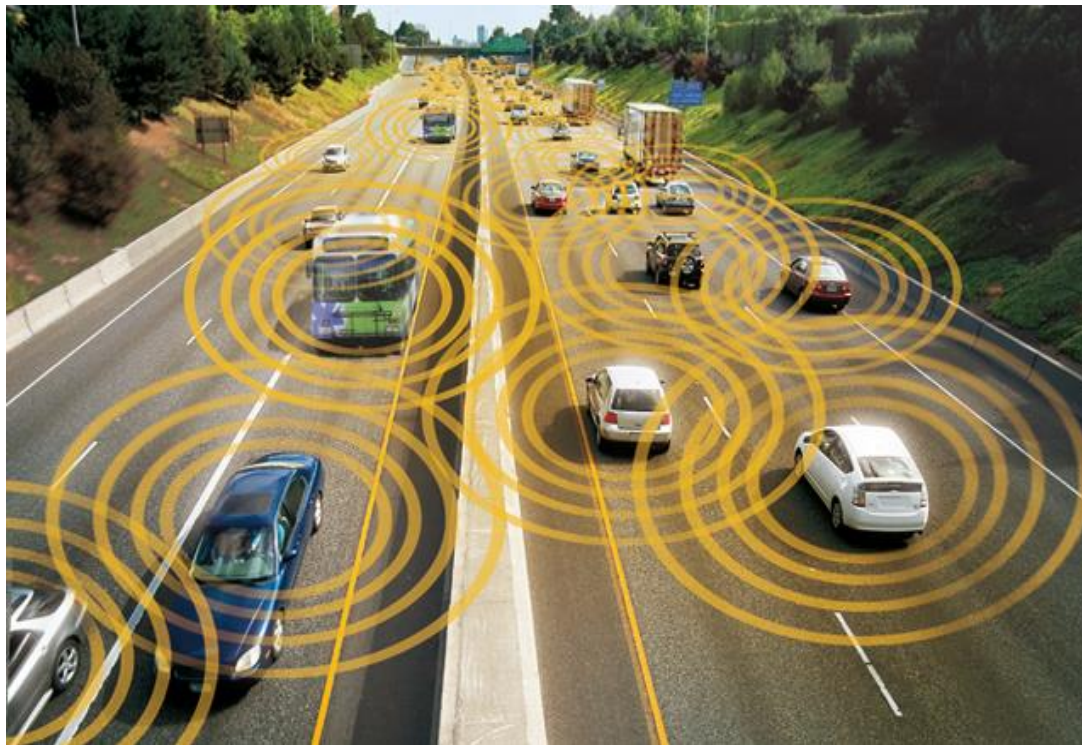
High neighbourhood density

- E.g., 30 vehicles in range (e.g., 1 km), 300 beacons/sec + event driven messages

Dynamically changing topologies

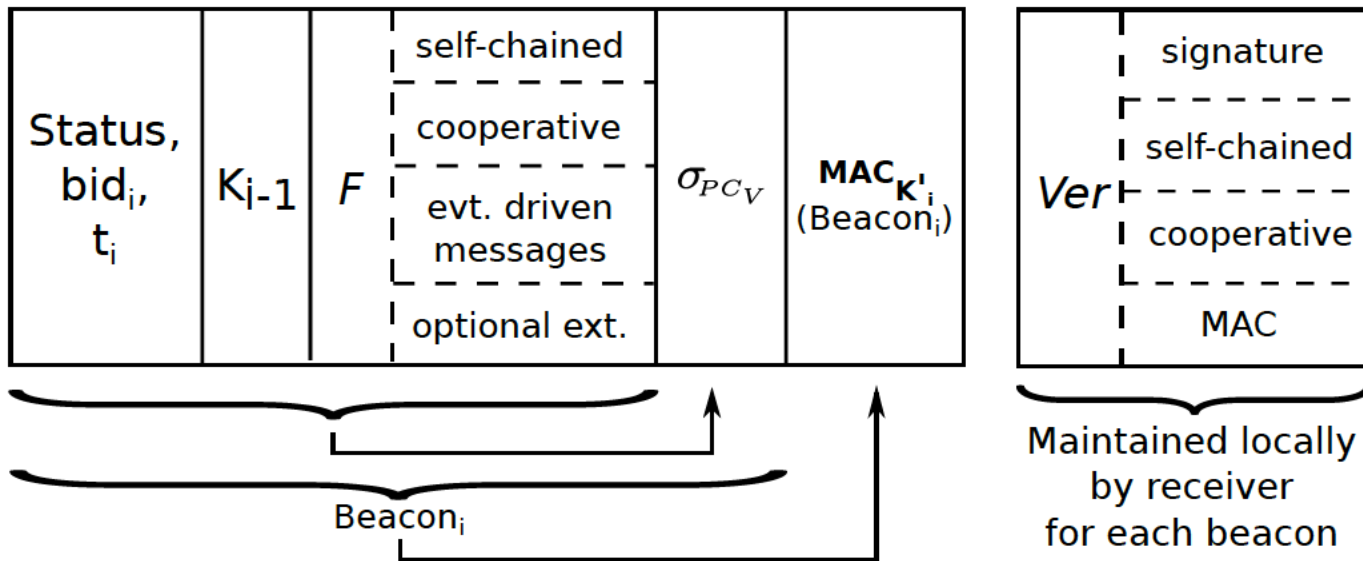
Attackers don't need to jam but keep connected vehicles too busy verifying bogus messages

- Not much of a choice: need to validate messages before deciding 'authentic or fake'



Attack Detection/Mitigation

Thwarting clogging DoS against VCS (continue)



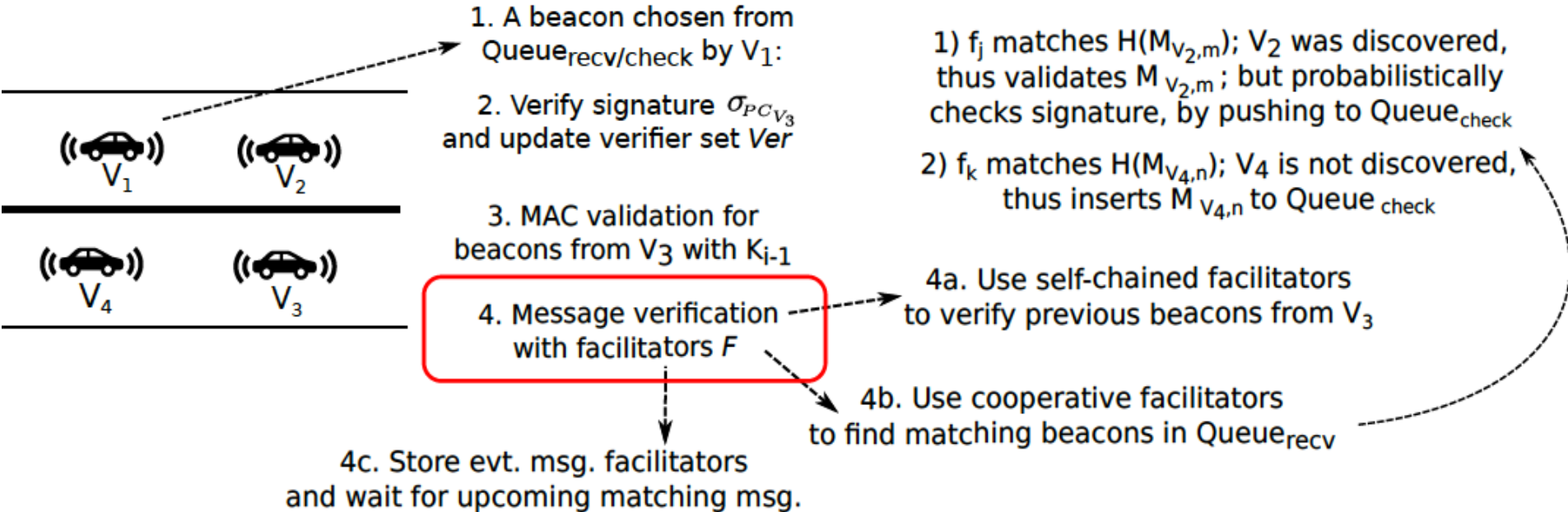
Solution

- Cooperation: vehicles facilitate validation at their peers, indicating which messages they validating

► Original beacon format: $\{Status, bid_i, t_i\}_{\sigma_{PCV}}$

Attack Detection/Mitigation

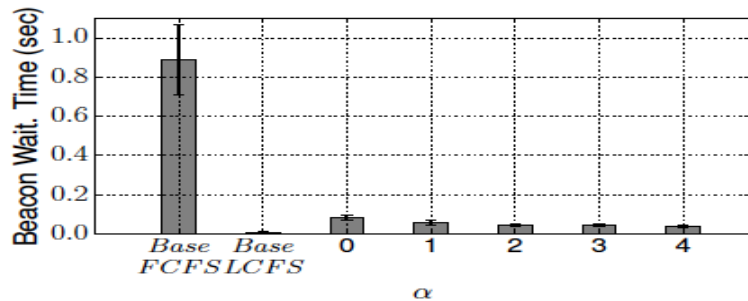
Thwarting clogging DoS against VCS (continue)



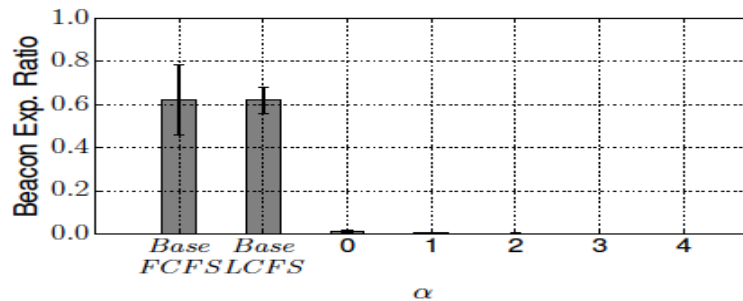
- Last-Come-First-Served (LCFS) $Queue_{recv}$ stores newly received beacon messages
- $Queue_{check}$ stores potentially valid beacons without cached PCs attached to them, and beacons chosen for signature verification (probabilistic checking and cross-checking)

Attack Detection/Mitigation

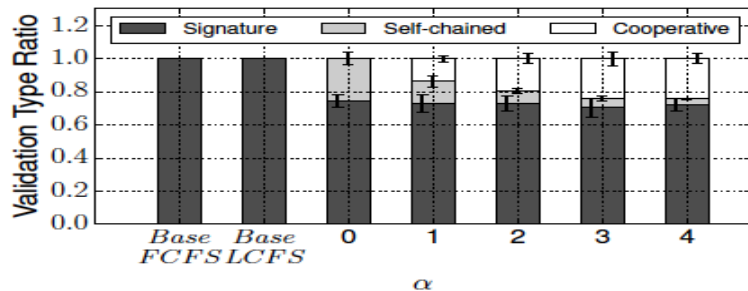
Thwarting clogging DoS against VCS (continue)



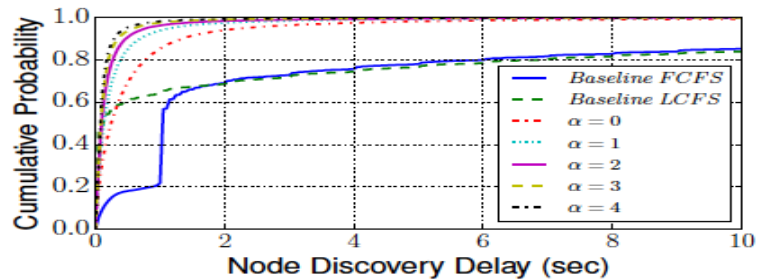
(a)



(b)



(c)



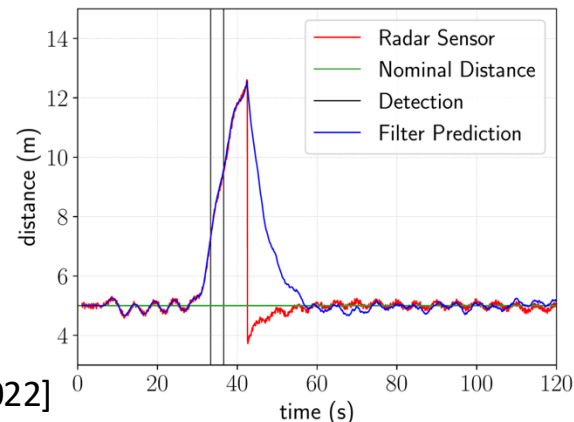
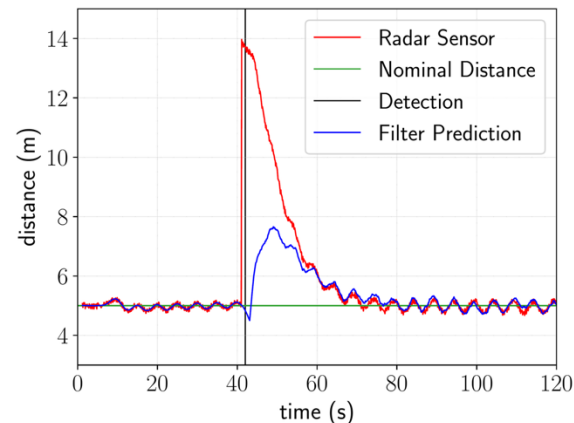
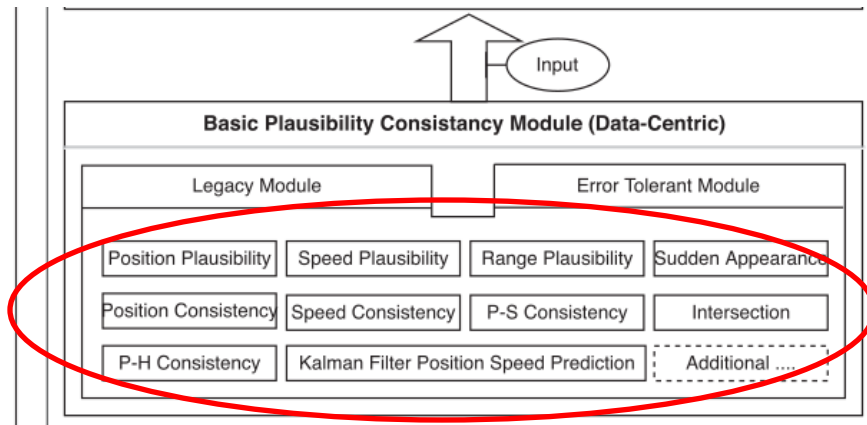
(d)

Figure 6: Beacon validation metrics as a function of α under the DoS attack.

Attack Detection/Mitigation

Basic/naive approaches

- Plausibility checks
 - Bypassed by smart attacks, e.g., gradual position increase
- Prediction models
 - Must be able to handle special cases
 - Manoeuvres are treated as attacks otherwise

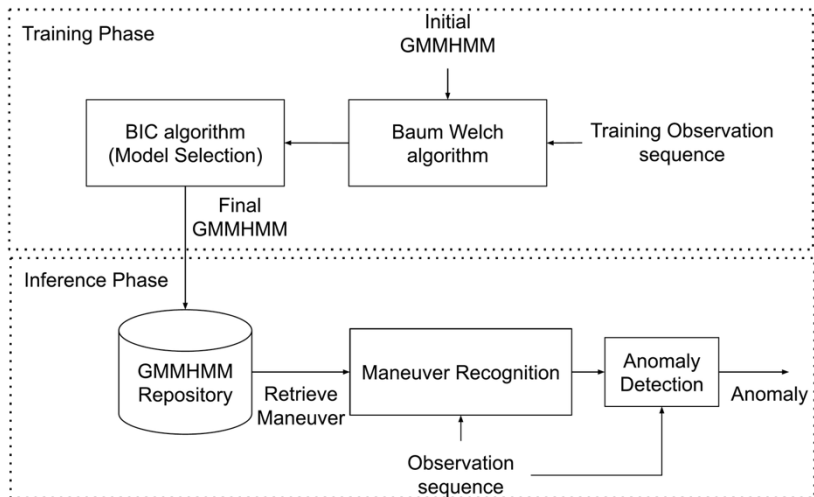


[KamelAPK:J:2020]

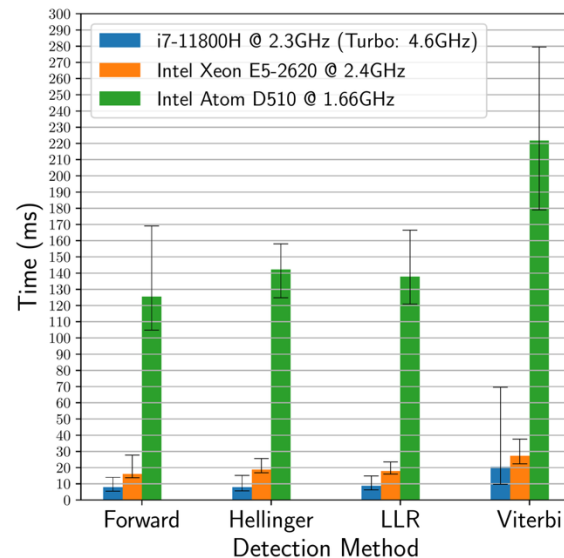
[KalogiannisKWP:C:2022]

Attack Detection/Mitigation

Learning approaches & Feasibility



- Ability to learn/train
 - Markov Models (HMM)
 - Supervised Learning
 - (Recurrent) Neural networks

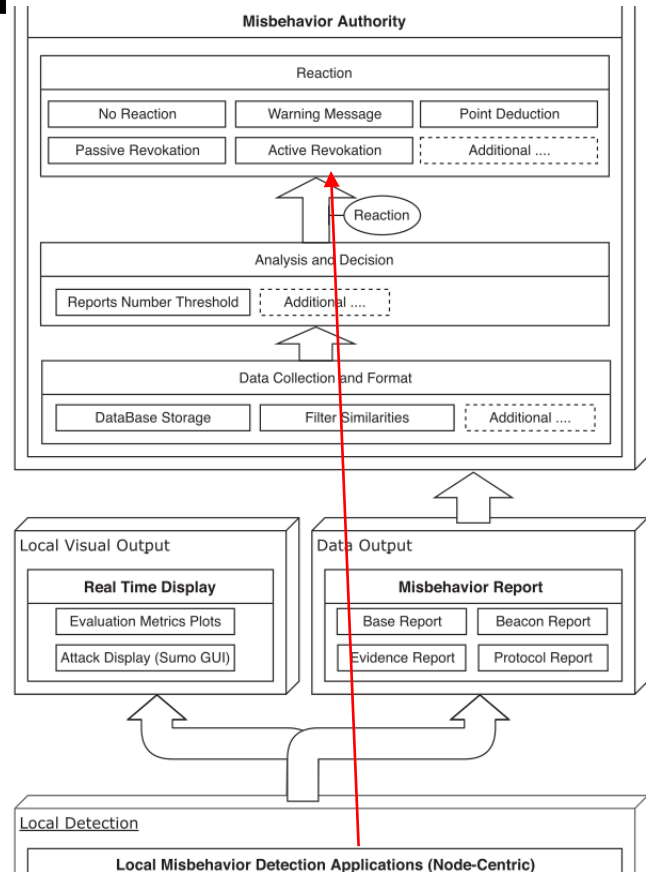


- Ability to reach conclusion fast
 - Remember: *Safety applications!*
 - Late processed beacons can be detrimental

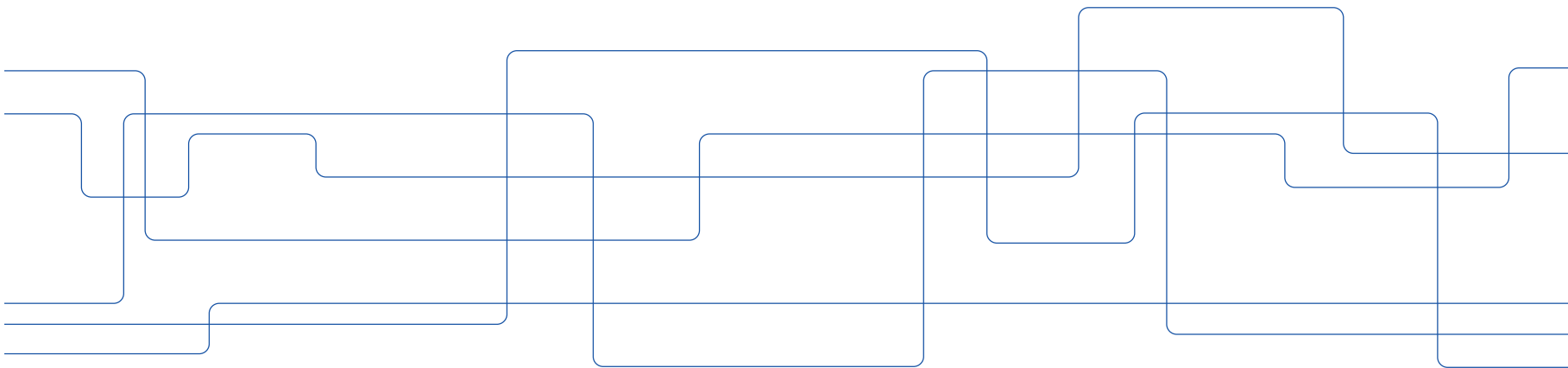
[KalogiannisKWP:C:2022]

Blame attribution & Exclusion

- Misbehaviour is detected locally
 - E.g., a vehicle detected an anomaly
- Send report to global authority
 - E.g., pseudonym X send Y violating Z
- Analyse report
 - Use past data
 - Use more power-consuming models
- Take action
 - E.g., revoke certificate

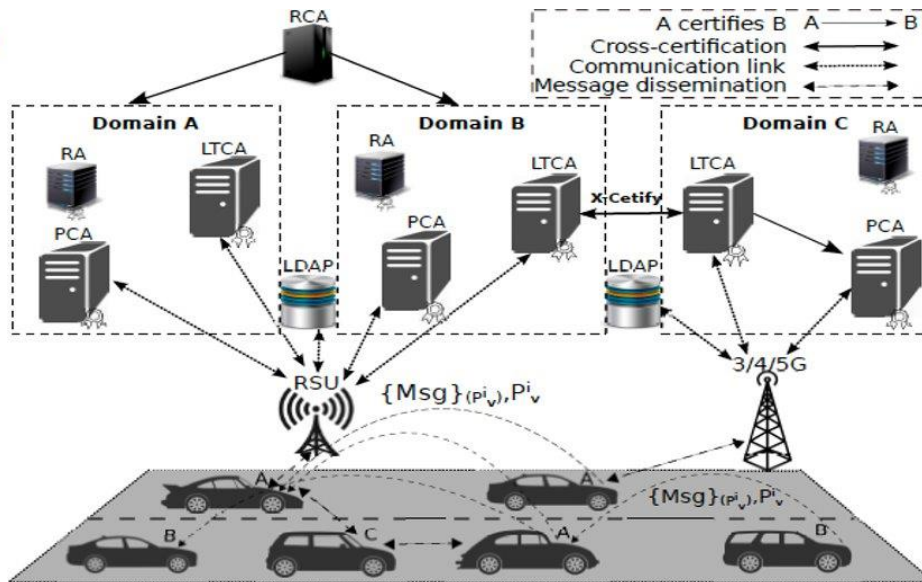


Scalable Vehicular Public Key Infrastructure (VPKI)



VPKI

- Vehicular Public-Key Infrastructure (VPKI)
- Root CA (RCA)
- Long Term CA (LTCA)
- Pseudonym CA (PCA)
- Resolution Authority (RA)
- Lightweight Directory Access Protocol (LDAP)
- Roadside Unit (RSU)



- Vehicles registered with one LTCA (home domain), one or more PCA servers per domains
- Vehicles can obtain pseudonyms from any PCA (home or foreign domains)
- RCA or cross-certification, de-anonymize (resolve pseudonyms) with the help of an RA

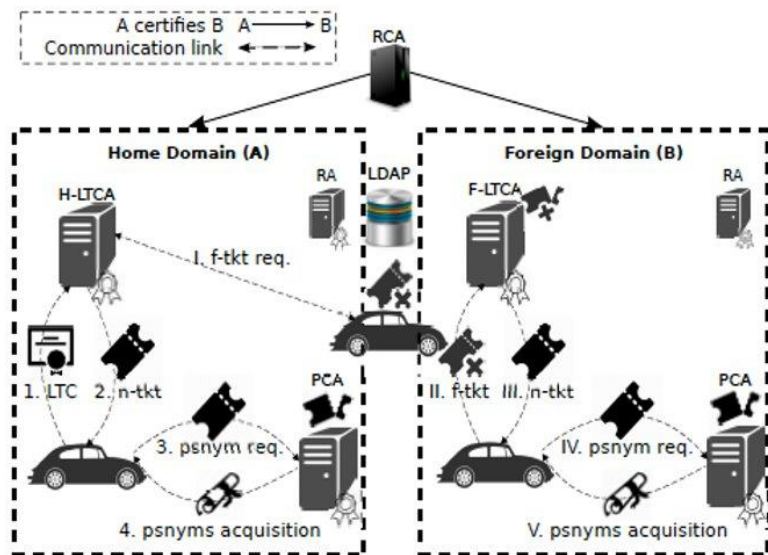
[KhodaeiJP:J:2018]



VPKI challenge

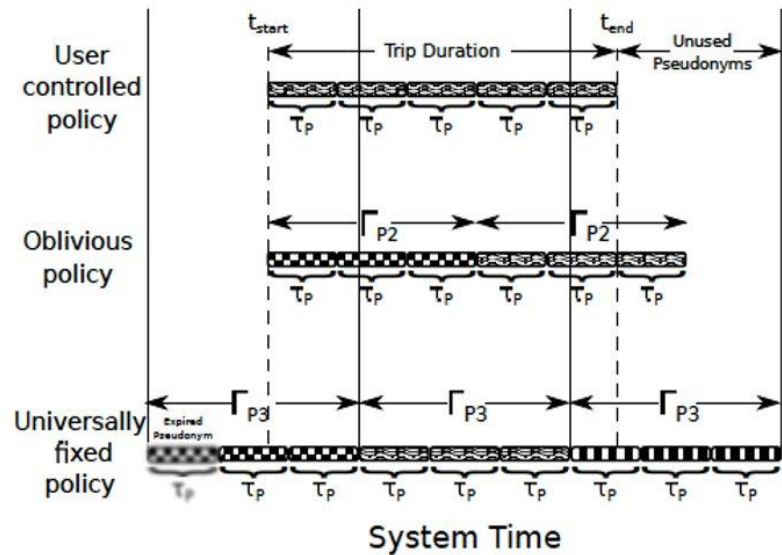
- **Traditional PKI vs. Vehicular PKI**
- Dimensions (5 orders of magnitude more credentials)
- Complexity and constraints
 - Balancing act: security, privacy, and efficiency
 - > *Honest-but-curious VPKI entities*
 - > *Performance constraints: safety- and time-critical operations (rates of 10 safety beacons per second)*
 - Multiple and diverse entities, global deployment, long-lived entities
 - Cost-driven platform resource constraints
- Mechanics of revocation
 - Highly dynamic environment
 - Short-lived pseudonyms, multiple per entity
 - Need for efficient and timely distribution of Certificate Revocation Lists (CRLs)
 - Strong privacy protection prior to revocation events

Scalable VPKI



Pseudonym acquisition overview in the home and foreign domains.

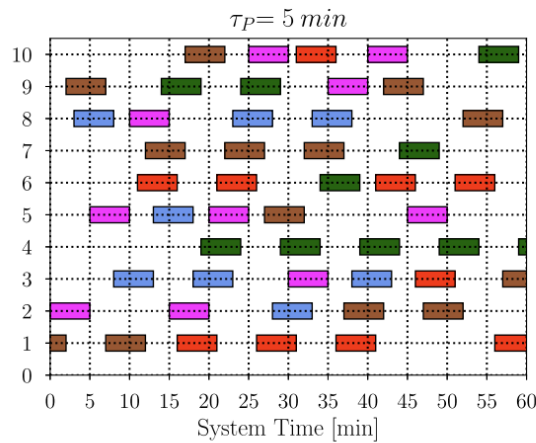
[KhodaeiJP:J:2018]



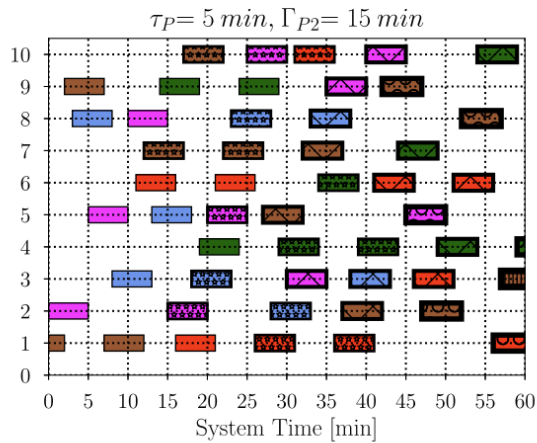
Pseudonym Acquisition Policies.

Scalable VPKI (cont'd)

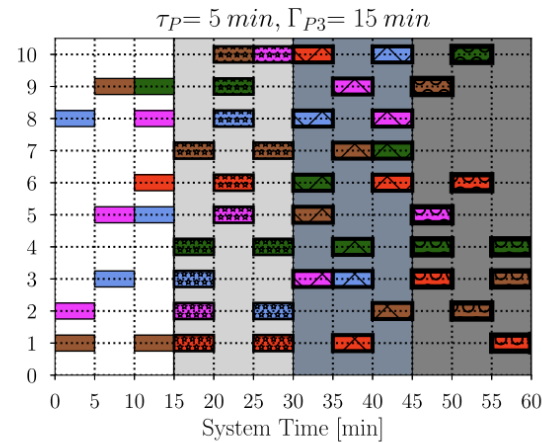
Pseudonym acquisition and change policies



(a)



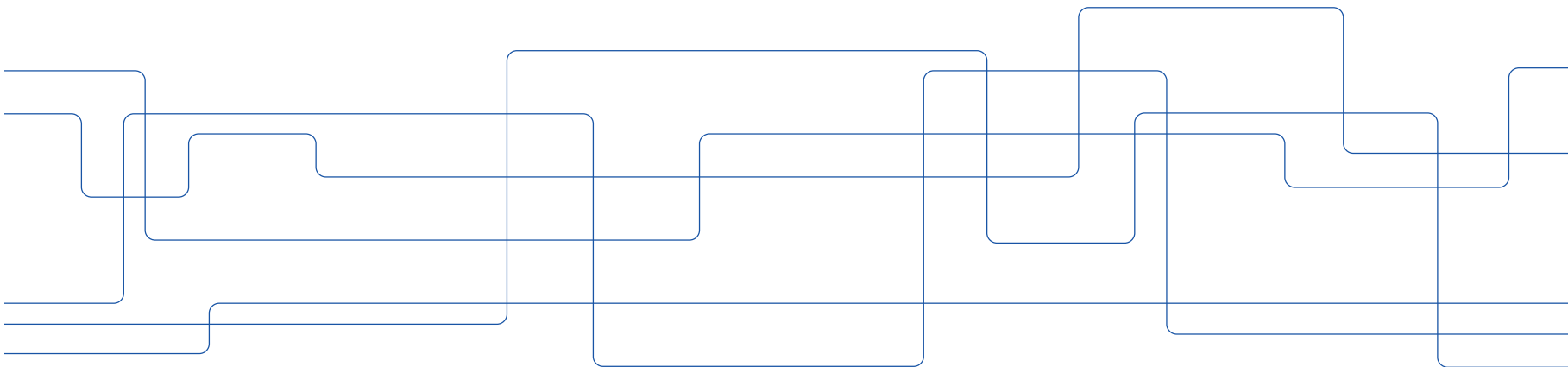
(b)



(c)

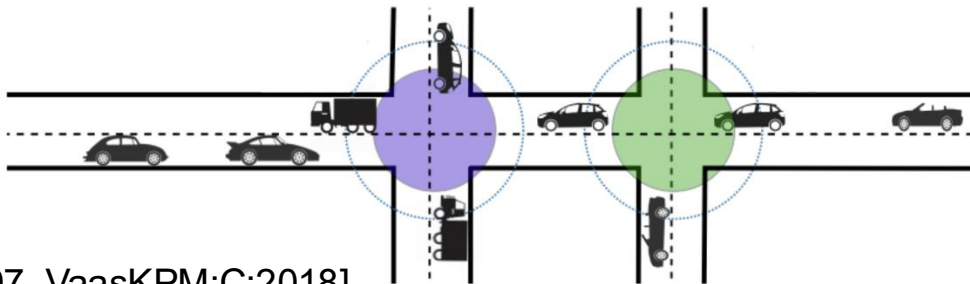
- (a) P1: User-controlled policy. (b) P2: Oblivious policy. (c) P3: Universally fixed policy.
- Each color shows the non-overlapping pseudonym lifetime
- Possible to infer the pseudonyms belong to the same vehicle based on pseudonym lifetime with P1 and P2

Privacy



Mix-zones for pseudonym change

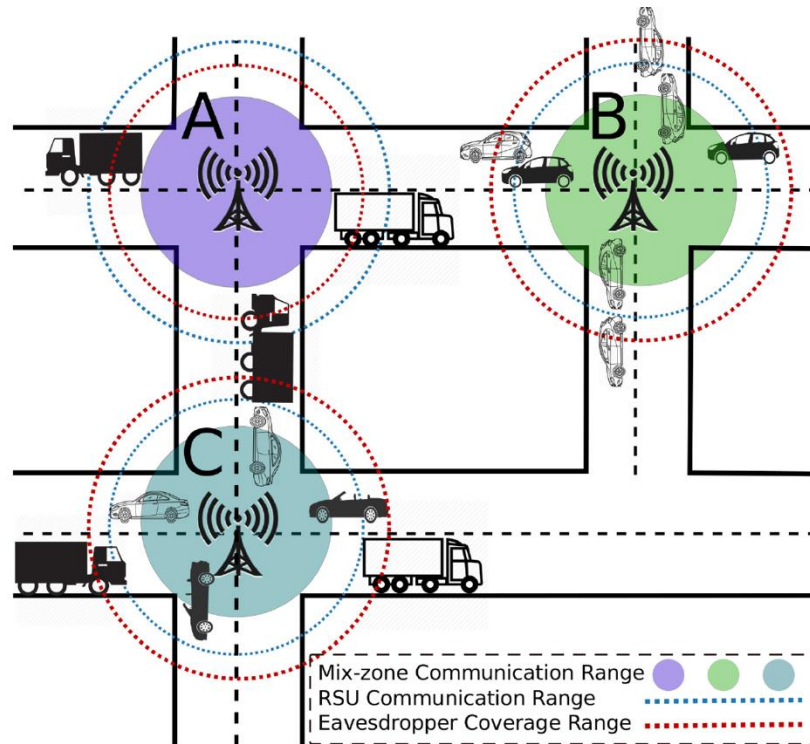
- Vehicles change pseudonyms in cryptographic mix-zones
 - Cryptographically protected messages in mix-zones
 - Hard to link the pseudonyms before and after the change by external eavesdroppers
- Challenge:
 - Physical constraints of the road layout
 - Mobility patterns (e.g., velocity, acceleration)
 - Vehicle density (e.g., sparse traffic conditions)



[FreudigerRFPH:C:2007, VaasKPM:C:2018]

Mix-zones with decoy traffic

- Chaff pseudonyms used to sign decoy traffic are obtained from the VPKI
- The VPKI proactively constructs a Cuckoo Filter (CF) by including chaff pseudonyms
- Legitimate vehicles efficiently filter out decoy traffic based on the signed CF



[KhodaeiP:J:2021a]

Mix-zones with decoy traffic (cont'd)

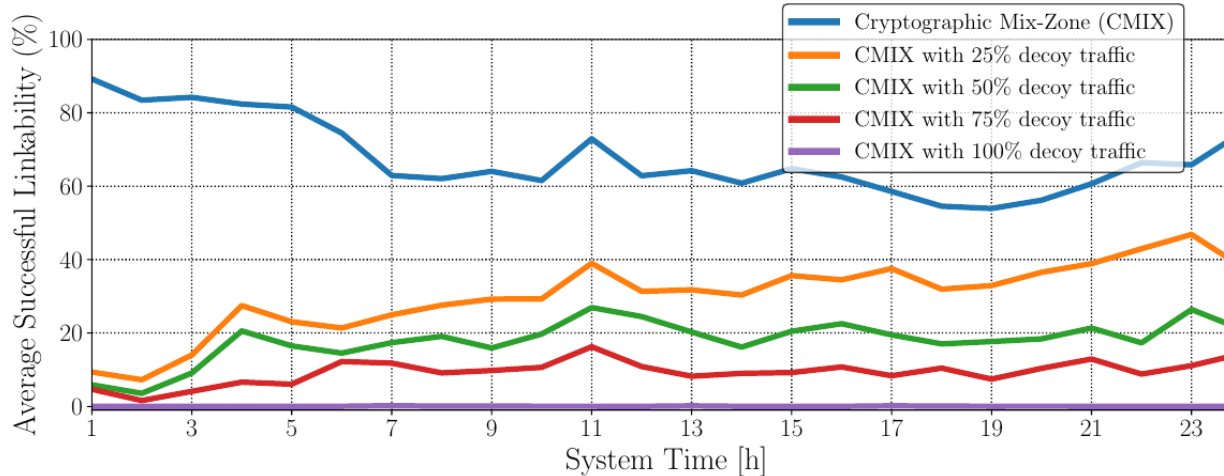
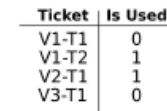


Figure: Syntactic and semantic linking attacks.

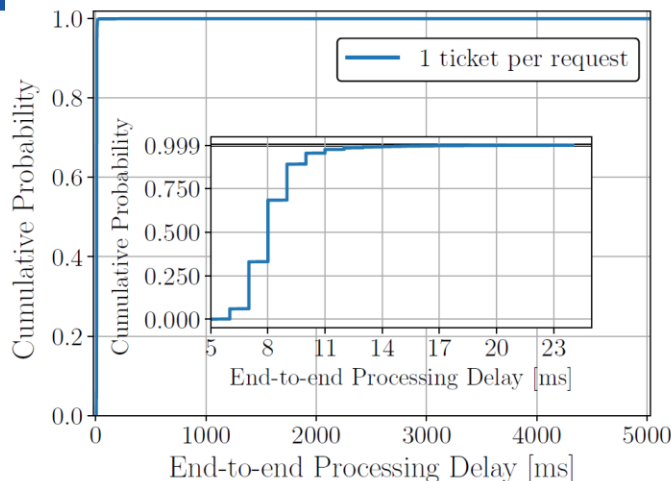
- The probability of linking decreases when the traffic density increases.
- The baseline scheme: successfully linking the pseudonyms with 78% success rates.
- With our scheme, the success rate decreases from $\approx 80\%$ to 40% by introducing decoy traffic (e.g., 25% of vehicles in the mix-zones).

- [KhodaeiNP:C:2019]

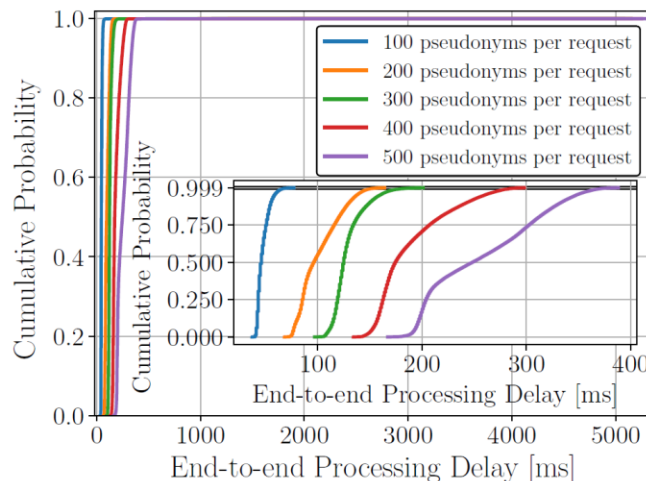


Certificate	Ticket Exp.
V1	T1
V2	T2
V3	T3
V4	T4

VPKlaaS (cont'd)



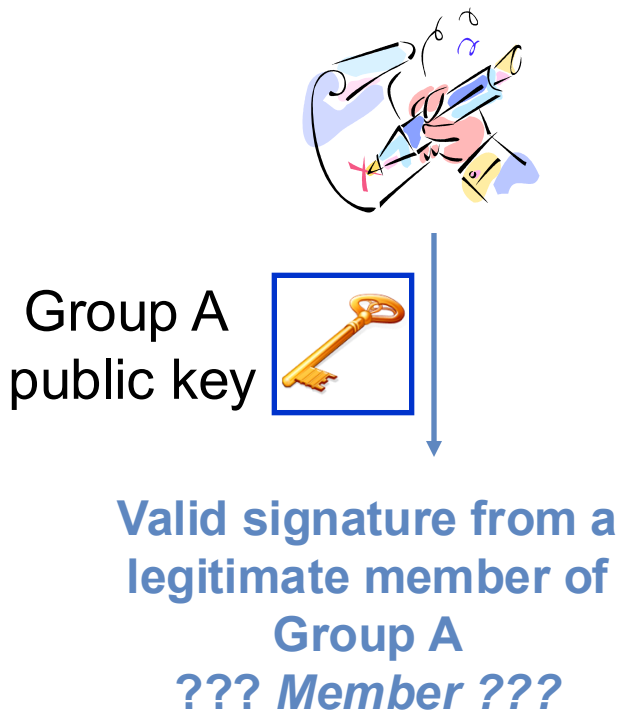
(a) E2E latency to issue a ticket



(b) E2E processing delay to issue psnyms

- End-to-end Latency for ticket: $F_x(t = 24 \text{ ms}) = 0.999$
- With a batch of 100 pseudonyms per request, 99.9% of the vehicles are served within less than 77 ms ($F_x(t = 77 \text{ ms}) = 0.999$)
- With a batch of 500 pseudonyms per request, the VPKlaaS system efficiently issues pseudonyms: $F_x(t = 388 \text{ ms}) = 0.999$

Group signatures



Group A



gsk_1



gsk_2



gsk_3

Hybrid scheme

- Combine
 - Pseudonymous authentication (Baseline Pseudonym (BP) approach) and
 - Group Signatures (GS)
- All legitimate vehicles belong to the same group
- Each node is equipped with a secret *group signing key* and the *group public key*
- Each node
 - Generates its own pseudonyms and signs them with a Group Signature
 - > GS act as a self-generated certificate
 - Uses the private key corresponding to the pseudonym to sign messages
 - > As in the baseline approach
 - Appends the self-generated certificate
 - > As in the baseline approach

[CalandrielloPHL:J:2011]

Hybrid scheme (cont'd)

Message formats

Baseline (BP)

m	$\sigma_{k_V^i}(m)$	K_V^i	$Cert_{CA}(K_V^i)$
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Group Signature (GS)

m	$\Sigma_{CA,V}(m)$
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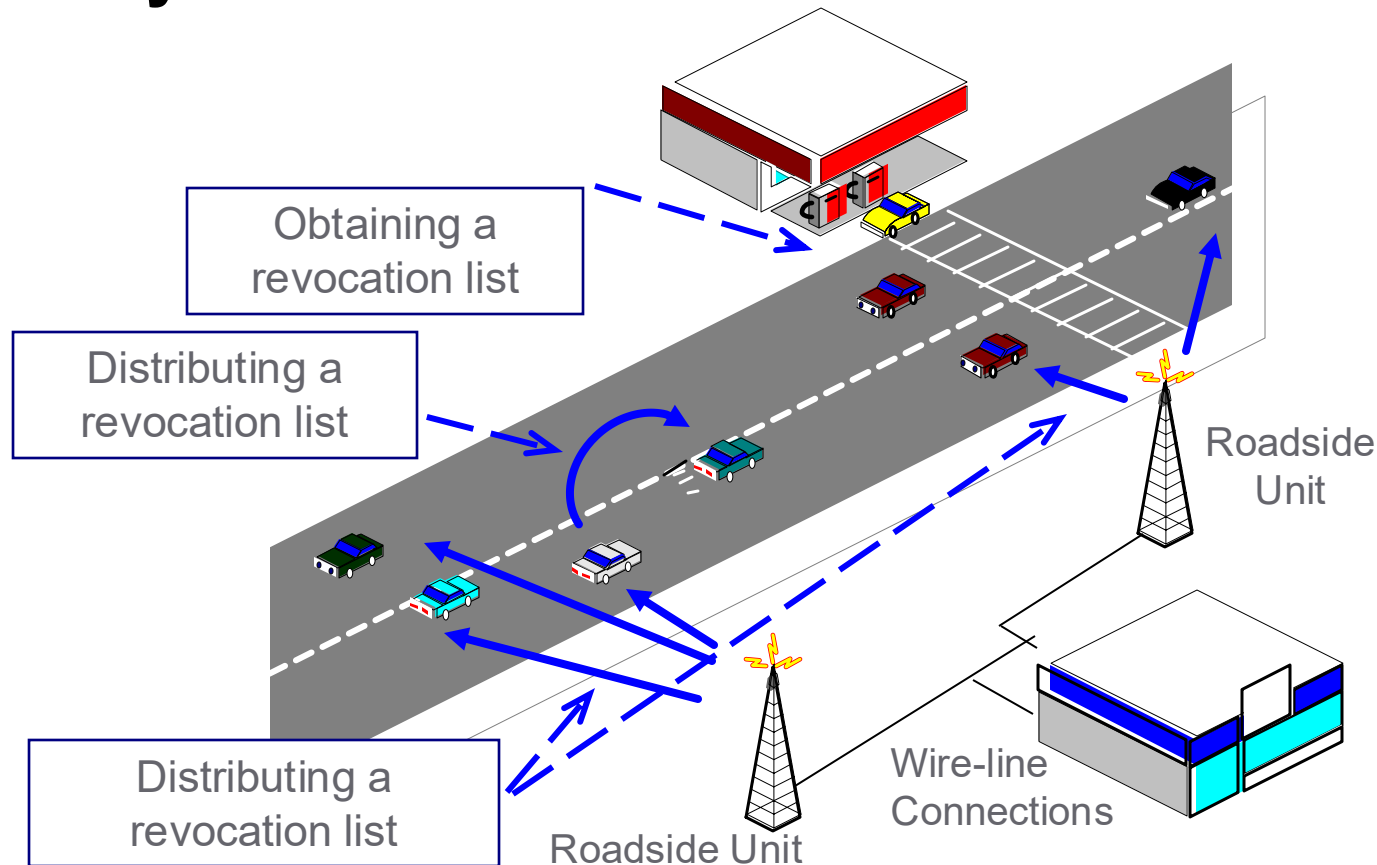
Hybrid

m	$\sigma_{k_V^i}(m)$	K_V^i	$\Sigma_{CA}^H(K_V^i)$
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Algorithm	Security level (bits)	Sign (ms)	Verify (ms)	Signature (bytes)	Public key (bytes)	Private key (bytes)
ECDSA-192	96	0.5	3	48	25	24
ECDSA-256	128	0.8	4.2	64	33	32
GS	128	53.7	49.3	225	800	64

Table: Computation costs and communication overhead for different signing algorithms

Faulty nodes eviction – certificate revocation

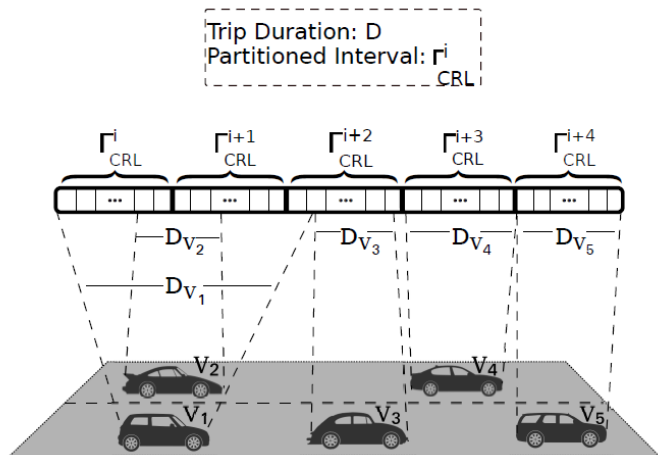




Certificate revocation

- Challenges
 - Scalability
 - Numbers of circulating and revoked vehicles
 - Multiple regions and Certification Authorities (CAs)
- Vehicular communication
 - Road-Side infrastructure Units (RSUs) are not pervasive
 - Short vehicle-RSU contact times
 - Bandwidth constraints
- Collaboration between domains (regional CAs)
 - Regional CRLs
 - Issuance of Foreigner Certificates

Vehicle-Centric CRL Distribution

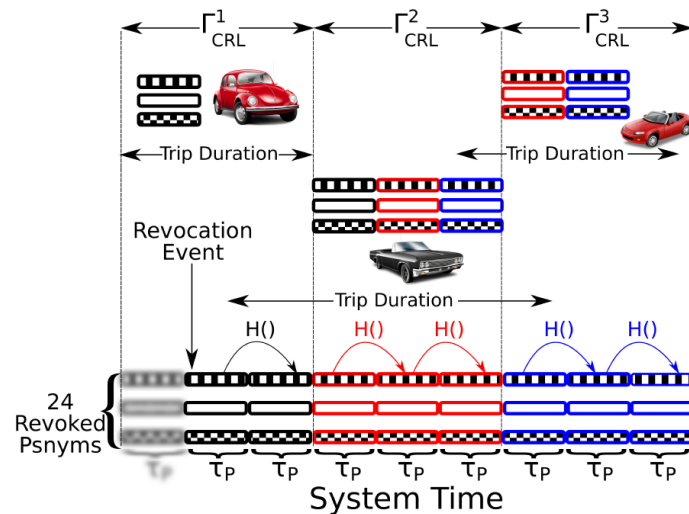


- CRL as a Stream:

V_1 subscribes to $\{\Gamma_{CRL}^i, \Gamma_{CRL}^{i+1}, \Gamma_{CRL}^{i+2}\}$;

$V_2 : \{\Gamma_{CRL}^i, \Gamma_{CRL}^{i+1}\}$;

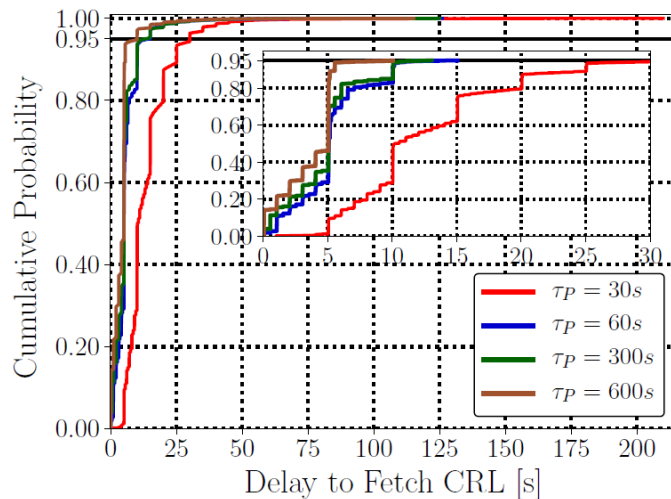
$V_3 : \{\Gamma_{CRL}^{i+2}\}$; $V_4 : \{\Gamma_{CRL}^{i+3}\}$; $V_5 : \{\Gamma_{CRL}^{i+4}\}$.



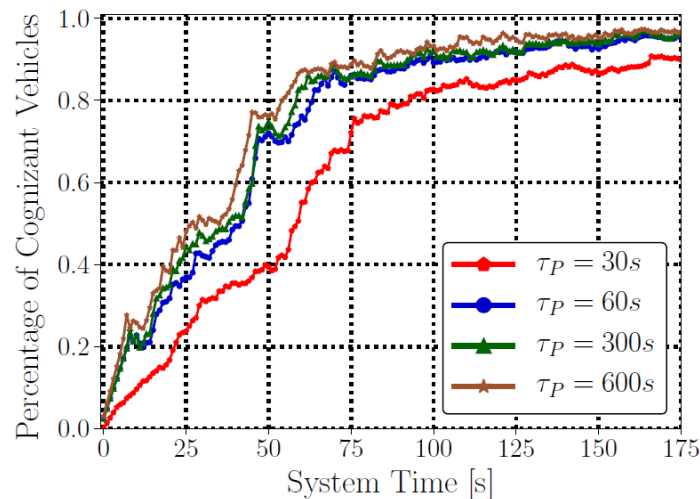
- A vehicle-centric approach: each vehicle only subscribes for pieces of CRLs corresponding to its trip duration.

[KhodaeiP:C:2018,KhodaeiP:J:2021b]

Vehicle-Centric CRL Distribution (cont'd)



(a) Vehicle-centric scheme ($\mathbb{B} = 10$ KB/s)



(b) Vehicle-centric scheme ($\mathbb{B} = 10$ KB/s)

Figure: (a) End-to-end latency to fetch CRL pieces. (b) Percentage of cognizant vehicles.

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